Rootopia: Depth-Resolved Root Trait Analysis for Minirhizotrons and Flatbed Scans in R

Purpose, Standard Workflow, and Extensions

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Introduction

How deep a root system reaches decides what it can get at. Water and nitrate move down through a profile and are lost from the top of it, phosphorus stays where it is, and carbon deposited below the plow layer persists longer than carbon deposited above it, so the same total root length means different things depending on the depth it occupies (Freschet *et al.* 2021). A trait reported as one number per image averages exactly that away.

Roots are buried, and the two standard ways of imaging them each give up something. A flatbed scanner captures washed roots at high resolution under controlled light, which gives morphology but destroys the plant and loses the depth each root grew at. A minirhizotron keeps both the plant and the depth, by imaging soil through the wall of a transparent tube – but it returns that tube's unrolled curved surface as a flat strip, in which pixel position is not depth. Depth has to be reconstructed from the tube's diameter, its insertion angle and the scan resolution before any trait can be attributed to a horizon.

Finding the roots in either image is a segmentation problem, and trained models solve it well enough to be used as given — RootDetector (Peters *et al.* 2023) and RootPainter (Smith *et al.* 2022) are two of them, and where the examples here came from. What they return is a mask. Turning that mask into root length density per 5 cm interval is not recognition but geometry, and it is the step at which a depth axis is either earned from measured quantities or quietly invented.

Rootopia performs that conversion and measures the traits that follow from it: root length, diameter, surface, branching architecture, network topology, turnover between sessions, and soil color and texture – each per depth interval wherever a depth map exists. The boundary matters in both directions:

- **Segmentation is somebody else's job.** `image_threshold()` is the one segmenter here, and it is the simplest kind: it splits foreground from background by brightness alone, in either direction. That is enough for washed roots on a tray of known color and not enough for roots in soil, where wet soil, old root and debris overlap in brightness. A trained model does that job, and handing over to one is a step in the workflow rather than an optional extra.
- **Depth is computed, not assumed.** It comes from the tube geometry and the scan resolution, which are numbers you supply; length estimators are named rather than chosen for you; and every function states which image type it expects.

This chapter covers the three image types and what consumes them, one complete worked depth profile, and each remaining module in turn. Part I is the standard minirhizotron workflow from stitched frames to a depth profile, Part II the modules that attach to it, and Part III the flatbed case, which has no depth axis and goes to architecture instead.

The two pathways

The package covers two imaging setups that share a front end and diverge after it. A minirhizotron sequence is stitched, segmented, and then reduced to traits per depth interval. A flatbed scan has no depth axis, so it goes to architecture instead: branch order, network shape, transport. Only stitching and segmentation are common to both, and segmentation happens outside this package.

Three things sit beside both paths rather than in either. **Root angle** needs a depth map and a skeleton together. **Soil color and texture** read the RGB scan and never touch root shape. **Turnover** compares two sessions. Part II takes each in turn.

Branch order belongs to the flatbed side. Nothing stops you running it on a tube scan, and `branch_order_map()` will return a graph – but a minirhizotron frame is a window onto a cylinder, so a root leaves and re-enters the field of view and the topology you recover is the topology of the visible fragments, not of the root system. Order and network metrics mean what they say on a washed root system laid flat.

On cleaning. `clean_image()` is optional on both sides but not neutral. `drop_objects_upto` removes connected components below a pixel count, which deletes scanner speckle and also deletes genuine fine-root fragments that enter and leave the frame – on a minirhizotron scan those fragments are real root. `fill_holes_upto` closes pinholes inside a mask, which raises area and lowers the measured diameter variance. Neither is reversible downstream, so the honest order is: measure once without cleaning, once with, and report which you used. For branch order the calculation is different – specks become spurious one-segment components and pinholes become spurious loops – so there cleaning earns its place.

`root_depth_metrics()` is not a step in either column. It is a batch wrapper that runs the whole left-hand path over a folder of images.

Two patterns cover everything downstream of the depth map, and both appear below. A trait that reduces to a per-zone sum or mean goes through `terra::zonal()` in one call. A trait whose function needs a whole image goes through `depth_zoning()`, which crops the image to one interval, in a loop.



Figure 1: The Rootopia hex logo, designed by Kaisa Jukuri.

Part I: the standard minirhizotron workflow

This is the path almost every minirhizotron analysis takes, in order: censor the rotation, build a depth map, bin it, measure traits per bin, summarize the profile. Nothing is thinned on the fly here: the example has a skeleton from the segmenter, so the hand-over described above is exercised as it stands. Where a segmenter returns no skeleton, one step is added and the section below shows what it costs. Part II then adds the modules that sit outside this path.

Stitch the frames, if there are several

Before anything else: a tube is often imaged as several overlapping frames, and a root crossing a frame boundary would otherwise be measured twice. If each tube is already one image, skip this. `stitch_root_scans()` groups files into tubes by a pattern in the filename, aligns consecutive frames over a band along the overlapping edge, and blends them into one mosaic per tube. Alignment is FFT phase correlation (`method = "phase"`); `method = "feature"` is accepted but stops with a message, because keypoint matching has no backend here.

Two arguments decide which files are read and how they are grouped, and they do different jobs. `pattern` is a plain substring that keeps or drops whole files; `group_regex` is the expression that pulls the tube identity out of each path. The default `group_regex` is already `"T0\\d{2}"`, which matches labels like `T067`, so a naming scheme built on `T0xx` needs nothing set. Anything else does: a folder named by plot and tube would use something like `group_regex = "P\\d+_T\\d+"`.

```
# See what a folder holds, then pick a range in the same call
stitch_root_scans("path/to/scans", tubes = "ask")

res <- stitch_root_scans(
  "path/to/scans",
  pattern     = ".tiff",         # read only these files
  group_regex = "T0\\d{2}",       # the tube identity inside each file name
  blend       = "max",           # see below: do not average a segmented mask
  order_by    = "\\d+(?=\\.)",    # sort frames on their number, not the file name
  report      = TRUE
)
res$report      # tube | step | dx | dy | peak | overlap
```

The report has one row per join. `peak` is how strongly the two edge bands agree at the shift that was chosen: a normalized correlation score, near 1 when the overlap matched cleanly and near 0 when the alignment is

little better than a guess. It is the column to check, because a bad join does not announce itself – the mosaic is still produced, with a seam in the wrong place.

```
# The ten least trustworthy joins in the run, worst first.
res$report[order(res$report$peak), ][1:10, ]
```

If a mosaic looks wrong, three arguments are worth changing before anything else. `edge_width` is the width of the band searched for the overlap, not the overlap itself, so it must be at least as wide as the true overlap – an overlap wider than the band cannot be found at all. `preprocess` conditions the bands before correlation: `"hann"`, `"grad"` or `"center"` help on scans with uneven lighting or sparse texture, where plain intensity correlation locks onto the illumination gradient instead of the roots. And `min_overlap` / `max_overlap` reject a shift that is physically impossible for your setup, which is the cheapest way to stop one bad join from propagating.

`blend` decides how the overlap is combined, and the choice is not cosmetic. `"linear"` feathers the two frames with an alpha ramp, which suits color scans. For a segmented mask use `"max"`: feathering *averages*, so a root pixel in one frame and background in the other becomes a fractional value, and thin roots ghost wherever alignment is off by a pixel. `"max"` takes the union instead, so a pixel is root if either frame called it root. `"overlay"` and `"overlay_first"` let one frame hide the other outright, and `"min"` takes the intersection, which erodes rather than ghosts.

Hand the images to a segmenter

Stitched or not, the next step is not in this package. A scan has to be turned into a root mask. Any tool that writes a binary mask will do; two that are used on minirhizotron images are:

- **RootDetector** (Peters *et al.* 2023) — a convolutional network. Returns the segmentation and a skeleton in one pass, and optionally the decay/production/constant coding that `root_turnover(method = "dpc")` reads.
- **RootPainter** (Smith *et al.* 2022) — interactive training, where you correct the model as it learns. Returns a mask, and no skeleton.

Those are examples, not requirements, and nothing below depends on either. Rootopia reads PNG, JPEG, TIFF, BMP, `terra` and `raster` objects, `imager` images and plain arrays, so the format is rarely the obstacle. What matters is which of the image types below you get back, not which program produced them.

Only one case does not need a trained model: a scan where root and background differ in brightness alone. `image_threshold()` compares each pixel against either a fraction of the image maximum (`method = "global"`) or a multiple of a local mean (`method = "adaptive"`). By default it keeps the pixels at or above that level, so the **bright** side is the foreground, which fits roots on a dark tray. A flatbed scan is normally the reverse, dark roots on a bright tray, and `invert = TRUE` moves the cut to the other side of the pixel value:

```
# Dark roots on a bright tray.
image_threshold(scan, threshold = 0.4, method = "global",
                binary_01 = TRUE, invert = TRUE)

# The adaptive form then asks that a pixel be adaptive_k times darker than its
# surroundings, rather than that much brighter.
image_threshold(scan, method = "adaptive", adaptive_k = 1.15,
                window_size = 51, binary_01 = TRUE, invert = TRUE)
```

This works only where brightness alone separates root from background, which in practice means washed roots on a tray of known color. It does not work on a minirhizotron scan: wet soil, old root and organic

debris overlap in brightness there, so no single cut and no local multiplier puts a boundary between them, whichever side you keep. That is a property of the image, not a shortcoming of the function, and it is why the hand-over to a trained segmenter is the step above rather than an optional extra.

What comes back is one or more image layers per scan. Which ones depends on the segmenter, and the three the rest of this article uses are described next.

The image layers, and which is which

A segmenter returns one or more image layers per scan, and they are not interchangeable. How many depends on the tool: a mask is the one you always get, a skeleton comes from some segmenters and not others, and the color photograph survives only if the setup recorded it. The three used below are the three a RootDetector run happens to produce, which is why the bundled example has all of them.

Figures here use `zoom_plot()`, as the step-by-step vignettes do. These rasters are several thousand pixels wide, and shrinking one to fit a figure makes the graphics device drop the one-pixel-wide roots entirely; `zoom_plot()` draws a full-image overview with the magnified region outlined, plus that region at native resolution where thin features survive. `frac` sets the inset size, so the magnification is `1 / frac`.

```
# Three layers from one RootDetector run on one tube; layer 2 carries the roots.
data(seg_Oulanka2023_Session03_T067)
data(skl_Oulanka2023_Session03_T067)
data(rgb_Oulanka2023_Session03_T067)

dpi <- 150   # the scan resolution; every cm value below depends on it

# Keep the full 3-band image (0/255) for the mask and depth steps below.
seg_img <- load_flexible_image(seg_Oulanka2023_Session03_T067,
                               output_format = "spatrast", scale = "none")

# load_flexible_image() also selects the root channel and binarizes it to 0/1
# in one call, so pixel sums and the void mask (abs(root_layer - 1)) are
# meaningful.
root_layer <- load_flexible_image(seg_Oulanka2023_Session03_T067,
                                  output_format = "spatrast",
                                  scale = "binary", select_layer = 2)

# The one-pixel center line, as this segmenter supplies it.
skl <- load_flexible_image(skl_Oulanka2023_Session03_T067,
                           output_format = "spatrast",
                           scale = "binary", select_layer = 2)

# The color functions read the photograph as it comes; drawing it wants 0-1.
# `scale` does both conversions, so neither is done by hand.
rgb      <- load_flexible_image(rgb_Oulanka2023_Session03_T067,
                                output_format = "spatrast", scale = "none")
rgb_disp <- load_flexible_image(rgb_Oulanka2023_Session03_T067,
                                output_format = "spatrast", scale = "to_01")
```

Each carries different information, so each feeds different functions.

Image	What it is	What it is for
RGB	the photograph	color, texture, soil class

Image	What it is	What it is for
Segmentation	solid root mask, 0/1	area, diameter, cleaning
Skeleton	one-pixel center line	length, tips, branching, root angle

Only the mask is guaranteed. Where no skeleton comes back, `skeletonize_image()` thins the mask into one, so a missing skeleton is never a blocker – but a supplied skeleton and a thinned one are not the same picture, and the section below compares them before anything downstream uses either.

Which image feeds which function

This is the table to keep. Passing the wrong type usually gives a wrong number rather than an error.

Function	Needs	Note
`count_pixels()`	segmentation	area, not length
`root_length()`	skeleton	`skeletonize = TRUE` thins a mask first
`branch_order_map()`	**both**	also where tips and forks come from
`root_diameter()`	**both**	mask for widths, skeleton for where
`network_metrics()`	segment table	from `branch_order_map()$edges`
`root_transport()`	segment table	needs diameters, so needs `mask =`
`root_scape_metrics()`	segmentation	patch structure of solid root (needs **landscapemetrics**)
`create_depthmap()`	any raster	reads the grid, not the values
`tube_coloration()`	RGB	three bands required
`classify_soil_rgb()`	RGB	per-pixel material class
`analyze_soil_texture()`	RGB	GLCM texture (needs **glcm**)
`root_turnover()`	two scans, or one DPC	per `tc_method`

The branching functions take an argument called `mask =`, and `root_diameter()` takes the same image as its first argument. In both cases what they want is the **segmentation**: the 0/1 image in which every root pixel is 1. The argument name is not a separate kind of image.

`root_diameter()` is the one to understand. Diameter comes from a distance transform of the segmentation: every root pixel is labeled with its distance to the nearest background pixel, which is the radius of the largest circle that fits inside the root there, and diameter is twice that. The skeleton then says *which* of those pixels to report, so each root contributes its center line once rather than its whole width. Pass only a skeleton and every diameter collapses to about one pixel. The same applies to `mask =` in the branching functions, where the continuation rule and the order-1 seed both depend on diameter.

`select_layer` applies to that first argument only, not to `skeleton_img`. It picks the band to read when the mask arrives as a multi-layer image; `NULL` passes every layer through, which is right when the mask is already one band, as `root_layer` is here.

Censor the rotation

A tube can turn between sessions, and the scanner does not cover a full circle, so roots near the edge of the scan are visible in one session and not the next. Left alone that is a systematic bias which looks like growth

or death. Cropping every session to a common window of rows removes it, and it is standard procedure rather than an option.

```
# The window has to be centered on the tube's rotational zero, and that is a
# per-scan quantity: estimate_rotation_center() locates it from the tape on the
# tube's upper side, and estimate_rotation_shift() measures how far a later
# session has turned relative to the first, which is what you add to the first
# session's center. Both need RStoolbox and imagefx, so the midpoint stands in
# here to keep this article runnable on a thin install.
r0 <- round(dim(root_layer)[1] / 2, 0)

censored <- rotation_censor(
  root_layer,
  center_offset  = r0,
  cut_buffer     = 0.02,
  fixed_width    = 900,    # rows kept; must fit inside the image height
  fixed_rotation = TRUE,   # equal width across sessions is what makes
  overlay        = TRUE,   # counts comparable
  main = "Rotation censor: kept (green) vs cut (red)"
)
```

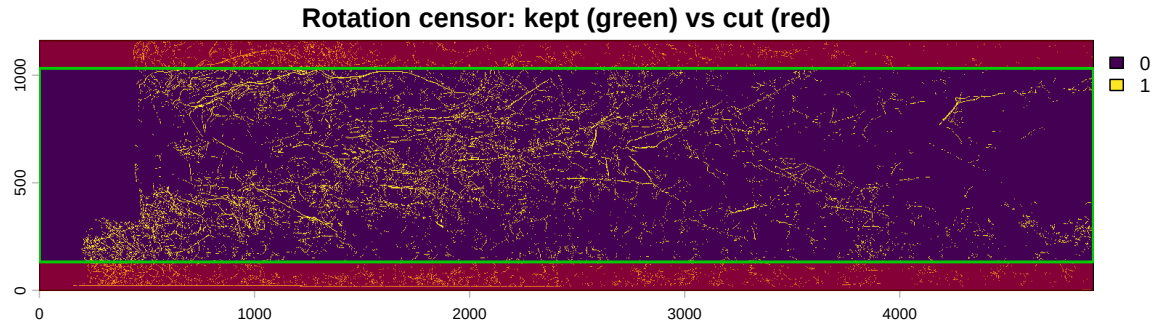


Figure 2: The retained window; the margins rotation leaves behind are discarded.

`fixed_rotation = TRUE` forces exactly `fixed_width` rows, which is what you want when comparing sessions: equal width is what makes counts comparable. `fixed_rotation = FALSE` instead cuts in proportion to a measured offset, so the output width varies between images. In a real run the order is: `estimate_rotation_center()` once per tube on the first session, `estimate_rotation_shift()` for every later session against it, and the sum passed as `center_offset` so the same band of tube is kept throughout. `center_offset` is a row index here; it is a fraction of the image height only in `create_depthmap()`, which is a different argument with the same name.

The rest of this article works on the uncensored image, so the depth figures cover the whole scan – which is still less than a full turn of the tube, since the scanner does not reach all the way around.

The skeleton: supplied, or made here

Some segmenters return a skeleton along with the segmentation; others return only the segmentation. Both cases work. Where a skeleton comes back, use it. Where it does not, `skeletonize_image()` makes one by stripping pixels off the edges of each root, layer by layer, until a single connected line is left down the middle.

There is no single right answer to that operation. Rootopia uses a Zhang-Suen *variant* — its lookup table differs from the published one at a handful of codes — and other tools use the published table, Guo-Hall, a medial axis transform, or something of their own. Each strips pixels in a slightly different order, so each leaves a slightly different line, most visibly at corners, at junctions and at the ends of short branches.

That matters because a skeleton drives length, tips and branching, so two numbers from the same segmentation can differ purely by which thinning produced the center line. And a segmenter that predicts the center line directly is not thinning anything at all: it is not bound by the segmentation it also predicts, so it can disagree with it.

The figure below overlays the two on one window: the segmenter's own skeleton against one thinned from the same segmentation. It is drawn from full-resolution originals rather than the bundled example, which is downsampled to keep the package installable. `data-raw/make-figures.R` writes it — set `seg_path` and it looks for the segmenter's skeleton beside it.

No comparison figure found. Set ***`seg_path`*** *at the top of* ***`data-raw/make-figures.R`*** *and run it.*

Neither line is the true one; they are two implementations of the same idea. What follows from that is only that the choice has to be recorded, and held fixed across a dataset — a supplied skeleton in one session and a thinned one in the next puts a step into a time series that has nothing to do with roots. Everything below uses the supplied skeleton.

Root diameter

Diameter needs both images at once: the mask gives the width, the skeleton says which pixels to report. `root_diameter()` returns the local thickness across the whole scan as well as the per-skeleton-pixel diameters, so it is worth looking at before reducing it to a per-interval mean.

```
diam_result <- root_diameter(
  root_layer,
  skeleton_img = skl,
  unit         = "cm",
  dpi          = dpi,    # not optional: the default is 300
  select_layer = NULL
)

# A window away from the center, where the roots are. zoom_plot() takes
# fractions of the extent, and row 1 is the top, so the y fraction counts up
# from the bottom.
zoom_plot(diam_result$distance_map_rast, frac = 0.1,
          center = c(1200 / terra::ncol(root_layer),
                     1000 / terra::nrow(root_layer)))
```

From pixels to depth

The depth map

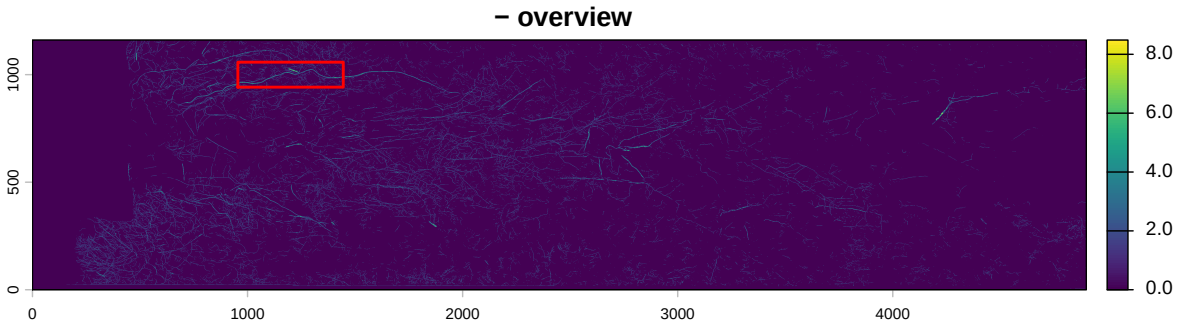


Figure 3: Local root thickness across the scan.

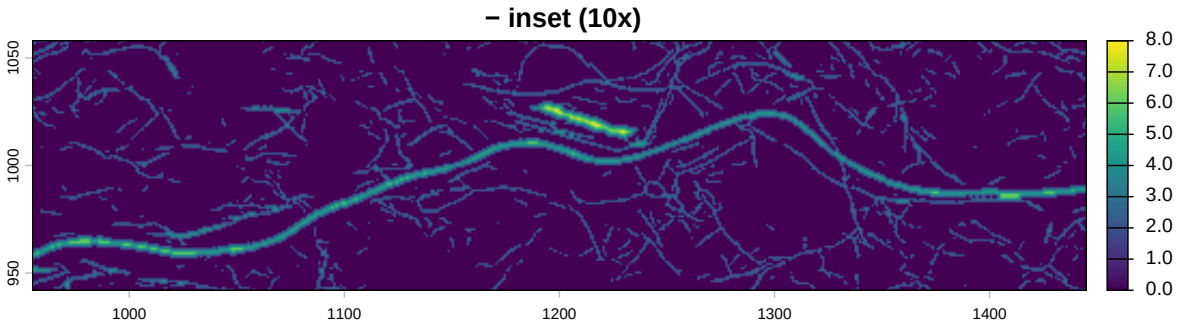


Figure 4: The same thickness map at native resolution.

```r
depth_map <- create_depthmap(
  img                 = root_layer,
  geometry            = "tube",  # "window" planar, "flat" no depth axis
  tube_thicc          = 7,       # cm - measure your own tube
  angle_from_vertical = 45,      # degrees from vertical; 0 is upright
  dpi                 = dpi,
  start_soil          = 2.9,     # cm - from in-situ calibration
  center_offset       = 0.5      # tube top halfway down the image
)

terra::plot(depth_map, main = "Depth map (cm)")
```

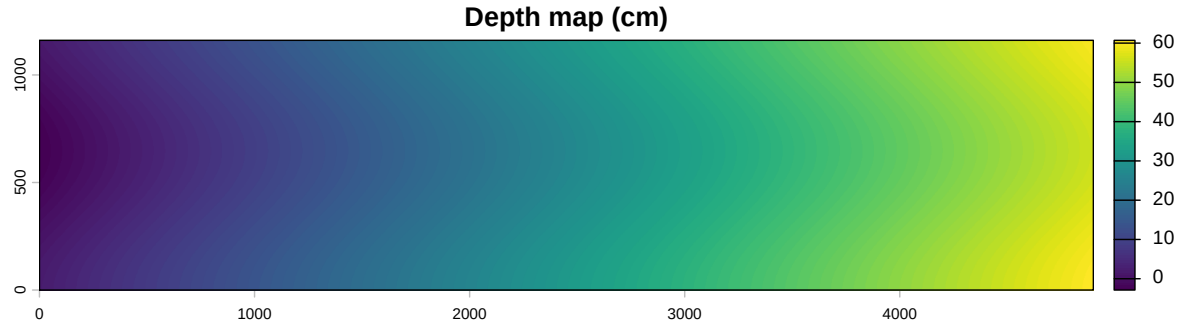


Figure 5: Depth along the tube, in centimeters.

Every argument above except `geometry` is a measurement, taken on the tube or read off the scanner, and each one fails in its own way if it is wrong.

- `angle_from_vertical` — the installation angle, in degrees from vertical, so 0 is an upright tube and 90 a horizontal one. From the field notes.
- `tube_thicc` — the inner diameter in centimeters, measured on the tube. If it or `dpi` is wrong the curvature wave cannot span the image height and the function stops with `Wrong Tube Diameter`, which is the one error here that catches itself.
- `dpi` — the scan resolution, from the scanner settings. Every centimeter in this article depends on it.
- `start_soil` — where the soil surface sits along the tube, in centimeters, from in-situ calibration. It is subtracted from every depth, so an error shifts the whole profile by that amount and changes nothing else. `estimate_soil_surface()` estimates it from the image in pixels; convert with `cm = px * 2.54 / dpi`.
- `center_offset` — where the top of the tube sits across the image height, as a fraction of one circumference, so 0.5 is halfway down. From the installation, or from `estimate_rotation_center()`.

Depth also runs along the image **width**. A scan with the soil surface at the top needs a quarter turn first; nothing errors if it is the wrong way round, the map is simply built along the wrong axis.

What the insertion angle does

A tube lying at an angle gains depth in two ways at once, and the insertion angle sets how much of each. Moving along the tube's length takes you deeper, and so does moving around its circumference from the top of the tube to the bottom. Standing the tube upright removes the first and leaves only the second; laying it flat does the reverse. Both terms are visible in the same map with `geometry = "tube"`.

```
# A 7 cm tube over 25 cm of its length: depth runs along the width, the
# circumference down the height, as in a real scan.
tube <- 7; sdpi <- 60
nr   <- round(tube * pi * sdpi / 2.54, 0) - 1          # one full turn
demo <- terra::rast(matrix(0, nrow = nr,
                           ncol = round(25 * sdpi / 2.54, 0)))

# Axes in centimeters rather than pixels, so the two directions can be read
# against each other.
terra::ext(demo) <- c(0, 25, 0, tube * pi)
angs <- c(15, 45, 75)

maps <- lapply(angs, function(a)
  create_depthmap(demo, geometry = "tube", tube_thicc = tube,
                  angle_from_vertical = a, dpi = sdpi, center_offset = 0))
lims <- range(vapply(maps, function(m) terra::minmax(m)[, 1], numeric(2)))

op <- graphics::par(mfrow = c(3, 1))
for (i in seq_along(angs))
  terra::plot(maps[[i]], range = lims, maxcell = Inf,
              xlab = "along the tube (cm)",
              ylab = "around the tube (cm)",
              main = sprintf("%d deg from vertical", angs[i]))
```

```
graphics::par(op)
```

Measuring the two terms off the function needs one more thing: a length of tube to measure the depth gain over. Taking 7 cm of tube, the same as the diameter, makes the two terms two halves of one budget.

```
budget <- terra::rast(matrix(0, nrow = nr,
                             ncol = round(tube * sdpi / 2.54, 0)))

ang_seq <- seq(0, 90, by = 5)
terms <- vapply(ang_seq, function(a) {
  m <- terra::as.matrix(
    create_depthmap(budget, geometry = "tube", tube_thicc = tube,
                    angle_from_vertical = a, dpi = sdpi, center_offset = 0),
    wide = TRUE)
  c(circ = diff(range(m[, 1])), len = diff(range(m[1, ])))
}, numeric(2))

op <- graphics::par(mar = c(4, 4, 2, 1))
plot(ang_seq, terms["circ", ], type = "l", lwd = 2, col = "coral3",
     ylim = c(0, max(terms)), xlab = "angle from vertical (deg)",
     ylab = "depth spanned (cm)")
graphics::lines(ang_seq, terms["len", ], lwd = 2, col = "steelblue4")
```

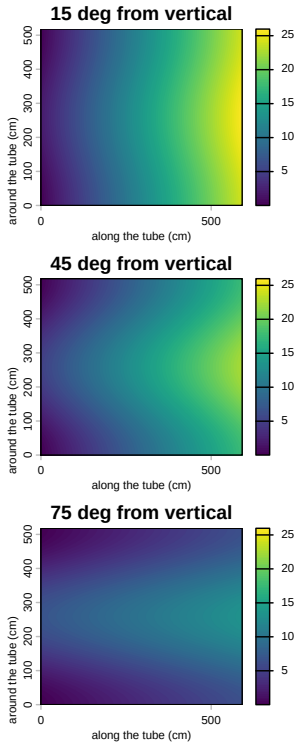


Figure 6: The same tube at insertion angles of 15, 45 and 75 degrees.

```
graphics::abline(v = 45, col = "grey70", lty = 3)
graphics::legend("top", bty = "n", lwd = 2, cex = 0.8,
                 col = c("steelblue4", "coral3"),
                 legend = c("along 7 cm of tube length",
                            "around a 7 cm tube"))
```

```
graphics::par(op)
```

Both curves are depth spanned in centimeters over the same 7 cm of tube: 7 cm around the circumference, 7 cm along the length. That makes them two halves of one budget. They cross at 45 degrees at the same value, 4.95 cm, and at either end one of them holds all 7 cm while the other holds none.

Near upright, almost all of the depth is the left-to-right gradient along the tube and the circumference sits at nearly one depth, so the banding is faint. Near horizontal it is the reverse: the banding spans a full tube diameter while the length gradient flattens. The two curves cross at 45 degrees, the angle most minirhizotron work uses and the one angle at which the two terms cannot be told apart. With `geometry = "window"` the circumference term is dropped and only the length term remains; with `"flat"` both go, and every pixel shares one depth.

Binning

```
depth_bins <- binning(depthmap = depth_map, nn = 5, round_option = "floor")

depths <- sort(unique(terra::values(depth_bins, mat = FALSE)))
```

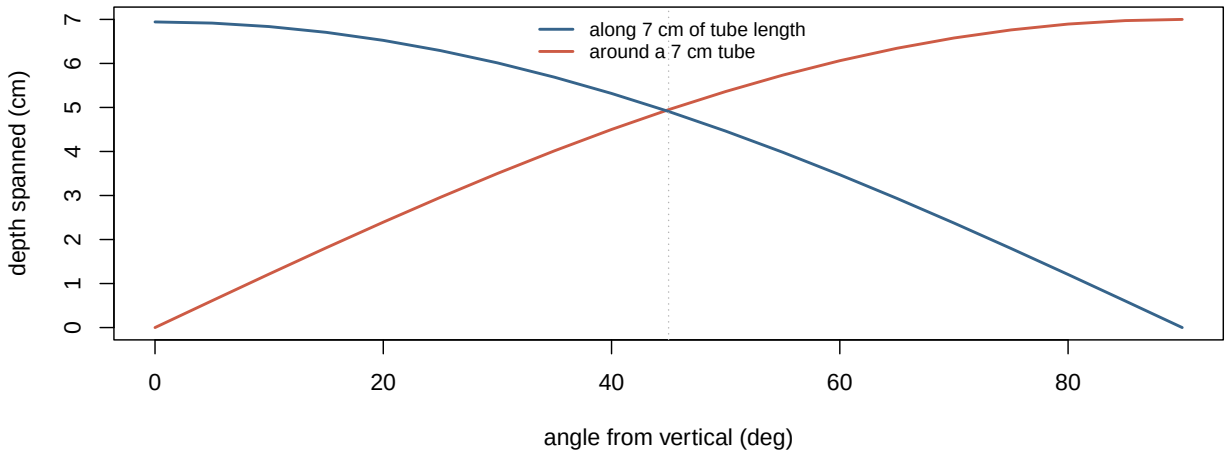


Figure 7: The two depth terms against insertion angle.

```
depths <- depths[!is.na(depths) & depths >= 0]
depths
#>  [1]  0  5 10 15 20 25 30 35 40 45 50 55 60
terra::plot(depth_bins, main = "Depth bins")
```

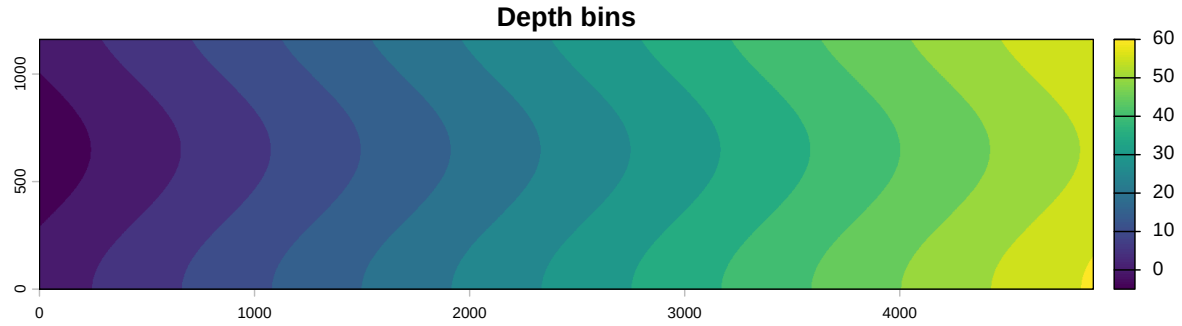


Figure 8: The depth map reduced to bins.

`round_option` decides what a bin's label means, and the two options label the same interval differently. `"floor"` computes `nn * floor(depth / nn)`, so a label is the **upper edge** of its interval: with `nn = 5` the bin labeled 5 holds depths 5 up to 10. The default `"rounding"` computes `nn * round(depth / nn)`, so a label is the interval's **center**: the bin labeled 5 holds 2.5 to 7.5, and the bin labeled 0 holds -2.5 to 2.5, which is half a bin wide once negative depths are dropped. Per-bin densities are unaffected either way, since they divide by each bin's own measured area, but any total that multiplies a per-bin value by `nn` is biased

by that half-width bin.

Traits per depth bin

Pixel counts reduce to a per-zone sum, so the whole profile comes out of one call.

```
root_px <- terra::zonal(root_layer, depth_bins, "sum", na.rm = TRUE)
void_px <- terra::zonal(abs(root_layer - 1), depth_bins, "sum", na.rm = TRUE)
colnames(root_px) <- c("depth", "rootpx")
colnames(void_px) <- c("depth", "voidpx")

depth_data <- merge(root_px, void_px, by = "depth")
depth_data <- depth_data[depth_data$depth >= 0, ]
depth_data$area_cm2 <- (depth_data$rootpx + depth_data$voidpx) *
                       (2.54 / dpi)^2
head(depth_data, 4)
#>   depth rootpx voidpx area_cm2
#> 2     0  10203 432006 126.7980
#> 3     5  58108 426762 139.0305
#> 4    10  50946 433824 139.0019
#> 5    15  47220 437614 139.0202
```

Root length cannot go through `zonal()` at all, for a plainer reason: `zonal()` applies a summary to the *values* in each zone, and length is not a property of pixel values but of how the pixels are arranged. `root_length()` has to see a connected skeleton. So the loop below does the same job `zonal()` does – one number per depth bin – with `depth_zoning()` supplying the crop `zonal()` would have supplied internally, and any function at all in place of a summary statistic.

One consequence of taking it bin by bin: Kimura's estimator weights diagonal and orthogonal steps differently, so it is not additive, and the per-bin lengths do not sum to the length of the whole skeleton. That is a property of the estimator rather than a leak, but it means a profile total and a whole-image total are different numbers and should not be compared.

```
length_by_depth <- do.call(rbind, lapply(depths, function(d) {
  skl_d <- depth_zoning(skl, depth_map = depth_bins, depth = d)
  data.frame(depth      = d,
             rootlength = root_length(skl_d, unit = "cm", dpi = dpi,
                                      show_messages = FALSE))
}))

depth_data <- merge(depth_data, length_by_depth, by = "depth")
depth_data$rootlength.density <- depth_data$rootlength / depth_data$area_cm2
head(depth_data, 4)
#>   depth rootpx voidpx area_cm2 rootlength rootlength.density
#> 1     0  10203 432006 126.7980   47.26035          0.3727215
#> 2     5  58108 426762 139.0305  232.97663          1.6757226
#> 3    10  50946 433824 139.0019  194.66974          1.4004829
#> 4    15  47220 437614 139.0202  187.38270          1.3478808
```

Diameter is a per-pixel value, so it reduces with `zonal()` like the counts, once `diam_result`'s raster is snapped onto the depth grid.

```r
terra::ext(diam_result$diameter_rast) <- terra::ext(depth_bins)

diam_by_depth <- terra::zonal(diam_result$diameter_rast, depth_bins,
                              "mean", na.rm = TRUE)
colnames(diam_by_depth) <- c("depth", "avg.diameter")

depth_data <- merge(depth_data, diam_by_depth, by = "depth")
head(depth_data, 4)
#>   depth rootpx voidpx area_cm2 rootlength rootlength.density avg.diameter
#> 1     0  10203 432006 126.7980   47.26035          0.3727215   0.03560018
#> 2     5  58108 426762 139.0305  232.97663          1.6757226   0.03746447
#> 3    10  50946 433824 139.0019  194.66974          1.4004829   0.03735308
#> 4    15  47220 437614 139.0202  187.38270          1.3478808   0.03611764
```

Profile summaries

```r
# Mean rooting depth - depth-weighted mean
ok      <- depth_data[!is.na(depth_data$rootlength.density) &
                      depth_data$rootlength.density > 0, ]
mrd_val <- MRD(w = ok$depth, roots = ok$rootlength.density)

# Total length is a sum of lengths. Summing a density and multiplying by the
# bin width is not a total of anything: density is cm of root per cm2 of
# window, so the product has no interpretation. The areal density of the whole
# profile is the total length over the total observed area.
total_len <- sum(depth_data$rootlength, na.rm = TRUE)
mean_rld  <- total_len / sum(depth_data$area_cm2, na.rm = TRUE)

cat(sprintf("MRD: %.2f cm | total length: %.1f cm | mean RLD: %.4f cm cm-2\n",
            mrd_val, total_len, mean_rld))
#> MRD: 20.64 cm | total length: 1226.7 cm | mean RLD: 0.7668 cm cm-2
```

`MRD()` stops on `NA`, on negative root values, and on an all-zero profile rather than returning a number built from part of the data, which is why the profile is filtered first and the filtering stays visible.

```r
library(ggplot2)

ggplot(depth_data, aes(x = depth, y = rootlength.density)) +
  geom_col(fill = "steelblue4") +
  coord_flip() +
  scale_x_reverse() +
  theme_minimal() +
  labs(x = "Soil depth (cm)", y = "Root length density (cm / cm2)",
       title = "Root length density profile")
```

```r
depth_data$Tube <- "T067"
depth_data$rootpx.cumulative <- root_accumulation(
  depth_data,
  group    = "Tube",
```

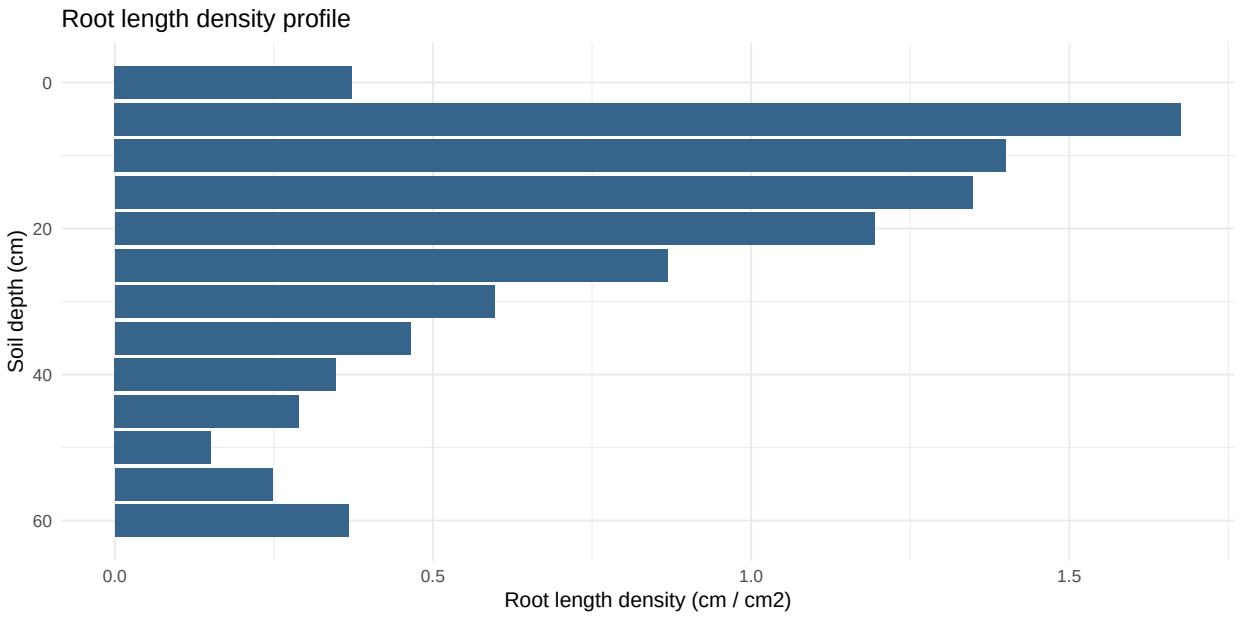


Figure 9: Root length density per depth bin.

```
  depth    = "depth",
  variable = "rootpx",
  stdrz    = "additive"    # counts: cumsum(x)
                           # additive: cumsum(x) / max(cumsum(x))
                           # relative: cumsum(x) / sum(cumsum(x))
)

ggplot(depth_data, aes(x = depth, y = rootpx.cumulative)) +
  geom_line(color = "steelblue4") +
  geom_point(color = "steelblue4") +
  theme_minimal() +
  labs(x = "Depth (cm)", y = "Cumulative root pixel fraction",
       title = "Root accumulation with depth")
```

All three options accumulate down the profile; `stdrz` only picks the denominator applied afterwards, and the names do not describe what they do. `"counts"` leaves the cumulative sum raw. `"additive"` divides it by its own maximum – that is, by the profile total – so the curve runs to 1 and tubes with very different abundance compare by shape alone. This is the one that gives a cumulative share, despite the name. `"relative"` divides by the *sum of the cumulative values*, a quantity with no obvious interpretation: the curve ends well below 1 and its endpoint depends on how many bins there are, so two profiles binned differently are not comparable. Use `"additive"` unless you have a specific reason, and read the documentation rather than the option name.

The whole workflow in one figure

Each step above produced a raster on the same grid, so they can be laid side by side: the scan, what the segmenter found in it, the center lines, the local thickness, the depth the geometry assigns to each pixel, and the profile those five produce. Depth runs along the image width, so a quarter turn puts it down the page and the panels stand next to each other.

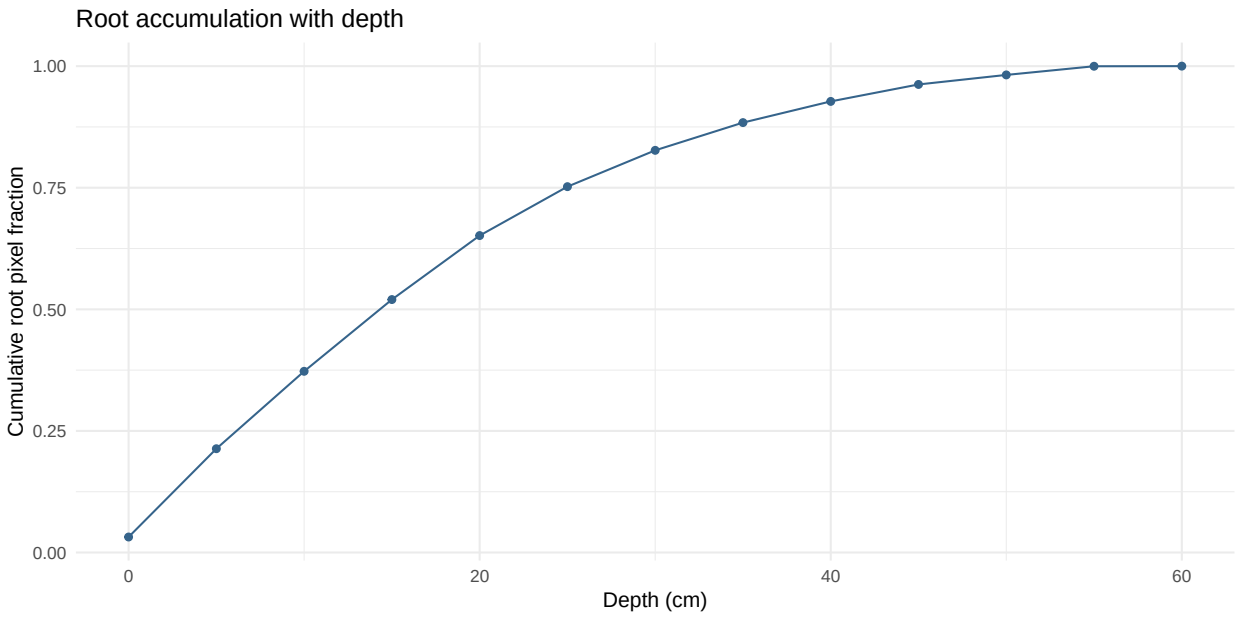


Figure 10: Cumulative root accumulation with depth.

```
# Order is left to right. Drop an entry to leave that panel out.
panels <- c("rgb", "root", "skel", "diam", "depth", "profile")

rot <- function(r) {
  r <- terra::flip(terra::t(r), direction = "horizontal")
  terra::ext(r) <- c(0, ncol(r), 0, nrow(r))
  r
}

# Layer 1 of the segmentation carries roots and foreign objects, layer 2 roots
# only, so a pixel in the first and not the second is tape.
seg_max   <- max(terra::global(seg_img, "max", na.rm = TRUE)[[1]])
is_masked <- (seg_img[[1]] / seg_max > 0.5) & (seg_img[[2]] / seg_max < 0.5)

# A gamma below 1 lifts the mid-tones without clipping the bright end.
root_soft <- root_layer^0.6

# distance_map_rast is the local diameter in pixels across the whole mask, so
# thickness already covers the root rather than sitting on its center line.
diam_mm <- diam_result$distance_map_rast * 2.54 / dpi * 10
diam_mm[diam_mm <= 0] <- NA

# Root area 1, center line 2.
skl_cls <- root_layer
terra::ext(skl_cls) <- terra::ext(skl)
skl_cls[skl == 1] <- 2

full <- list(rgb = rgb_disp^0.7, root = root_soft, skel = skl_cls,
             diam = diam_mm, depth = depth_bins)
for (nm in c("root", "skel", "diam")) full[[nm]][is_masked] <- NA
```

```r
full <- lapply(full, rot)

# Aggregating to about the on-page pixel count first means terra never drops
# pixels while drawing, which is what loses one-pixel-wide roots.
afun  <- c(rgb = "mean", root = "max", skel = "max", diam = "max", depth = "mean")
panel <- lapply(names(full), function(n)
  terra::aggregate(full[[n]], fact = 4, fun = afun[[n]], na.rm = TRUE))
#> |---------|---------|---------|---------|=========================================
names(panel) <- names(full)

area_ramp  <- grDevices::colorRampPalette(c("black", "#2E7DD1"))(100)
black_ramp <- c("black", "black")
skel_cols  <- c("black", "#2E7DD1", "#FF5A36")
diam_cols  <- grDevices::colorRampPalette(c("#f5e07a", "#80066b"),
                                          space = "Lab")(100)
band_cols  <- grDevices::colorRampPalette(
  c("#FFD92F", "#66BD63", "#00A6CA", "#2455A4"))(length(depths))
# Roots over the depth bands: transparent at 0, near-black at 1.
root_ramp  <- grDevices::rgb(0.07, 0.07, 0.07, alpha = seq(0, 1, length.out = 100))
leg_bg     <- grDevices::adjustcolor("black", 0.65)
mask_col   <- "grey85"
diam_rng   <- as.numeric(stats::quantile(terra::values(panel$diam, mat = FALSE),
                                         c(0.02, 0.98), na.rm = TRUE))
mar_p      <- c(0.4, 0.4, 1.8, 0.4)

# ---- zoom inset ----
zoom_on   <- TRUE
# c(across, down) as fractions of the panel. "densest" instead searches for the
# window holding the most skeleton pixels, which moves between scans; a fixed
# position is reproducible, and this one is the position used throughout.
zoom_ctr  <- c(0.8, 0.35)
zoom_w_default <- 0.10
# Window width per panel, as a fraction of the panel width. Magnification in a
# panel is inset_w / its entry, so each panel can be zoomed separately.
zoom_w    <- c(rgb = zoom_w_default, root = zoom_w_default,
               skel = zoom_w_default, diam = zoom_w_default,
               ang = 0.08, depth = zoom_w_default)
inset_w   <- 0.62
inset_pos <- "topright"     # topright, topleft, bottomright, bottomleft
zoom_col  <- "firebrick4"
zoom_lwd  <- 2

W <- ncol(full$root); H <- nrow(full$root)
zw  <- zoom_w_default * W
iw  <- inset_w * W
pad <- 0.03 * W
ix  <- if (grepl("right", inset_pos)) c(W - pad - iw, W - pad) else c(pad, pad + iw)
iy  <- if (grepl("top",   inset_pos)) c(H - pad - iw, H - pad) else c(pad, pad + iw)
inset_e <- terra::ext(ix[1], ix[2], iy[1], iy[2])

if (identical(zoom_ctr, "densest")) {
  # Count skeleton pixels in coarse blocks, sum them over a window-sized
  # neighborhood, and take the best window that does not sit under the inset.
```

```
  step <- 10
  dens <- terra::aggregate((full$skel == 2) * 1, fact = step, fun = "sum",
                           na.rm = TRUE)
  wk   <- 2 * round(zw / step / 2) + 1
  dens <- terra::focal(dens, w = wk, fun = "sum", na.rm = TRUE)
  xy   <- terra::xyFromCell(dens, seq_len(terra::ncell(dens)))
  v    <- terra::values(dens, mat = FALSE)
  under <- xy[, 1] > inset_e[1] - zw & xy[, 1] < inset_e[2] + zw &
           xy[, 2] > inset_e[3] - zw & xy[, 2] < inset_e[4] + zw
  edge  <- xy[, 1] < zw / 2 | xy[, 1] > W - zw / 2 |
           xy[, 2] < zw / 2 | xy[, 2] > H - zw / 2
  v[under | edge] <- NA
  best <- xy[which.max(v), ]
  zoom_ctr <- c(best[1] / W, 1 - best[2] / H)
}

# One square window per panel, all centered on zoom_ctr, moved inward if the
# window would hang over the panel edge.
make_win <- function(w) {
  h <- w * W / 2
  x <- min(max(zoom_ctr[1] * W,       h), W - h)
  y <- min(max((1 - zoom_ctr[2]) * H, h), H - h)
  terra::ext(x - h, x + h, y - h, y + h)
}
wins <- lapply(zoom_w, make_win)
win  <- wins$depth
cx   <- (win[1] + win[2]) / 2

# new = TRUE starts the inset on its black backdrop, FALSE adds a layer on top.
draw_inset <- function(r, win, new = TRUE, ...) {
  if (!zoom_on) return(invisible(NULL))
  if (new) graphics::rect(inset_e[1], inset_e[3], inset_e[2], inset_e[4],
                          col = "black", border = NA)
  w <- terra::crop(r, win); terra::ext(w) <- inset_e
  if (terra::nlyr(w) == 3) {
    terra::plotRGB(w, r = 1, g = 2, b = 3, scale = 1, add = TRUE,
                   axes = FALSE, maxcell = Inf)
  } else {
    terra::plot(w, add = TRUE, legend = FALSE, axes = FALSE, maxcell = Inf, ...)
  }
  graphics::rect(inset_e[1], inset_e[3], inset_e[2], inset_e[4],
                 border = zoom_col, lwd = zoom_lwd)
  graphics::rect(win[1], win[3], win[2], win[4],
                 border = zoom_col, lwd = zoom_lwd)
}

draw <- list()

draw$rgb <- function() {
  terra::plotRGB(panel$rgb, r = 1, g = 2, b = 3, scale = 1, axes = FALSE,
                 maxcell = Inf, mar = mar_p)
  draw_inset(full$rgb, win = wins$rgb)
  graphics::mtext("Scan", side = 3, line = 0.4)
```

```
}

draw$root <- function() {
  terra::plot(panel$root, col = area_ramp, colNA = mask_col, range = c(0, 1),
              legend = FALSE, axes = FALSE, maxcell = Inf, mar = mar_p)
  draw_inset(full$root, win = wins$root, col = area_ramp,
             colNA = mask_col, range = c(0, 1))
  graphics::mtext("Segmentation", side = 3, line = 0.4)
}

draw$skel <- function() {
  terra::plot(panel$skel, breaks = c(-0.5, 0.5, 1.5, 2.5), col = skel_cols,
              colNA = mask_col, legend = FALSE, axes = FALSE, maxcell = Inf,
              mar = mar_p)
  draw_inset(full$skel, win = wins$skel, breaks = c(-0.5, 0.5, 1.5, 2.5),
             col = skel_cols, colNA = mask_col)
  graphics::legend("bottomleft", legend = c("root", "skeleton"), bty = "n",
                   fill = skel_cols[2:3], border = NA, bg = leg_bg,
                   text.col = "white", cex = 0.65, inset = 0.02)
  graphics::mtext("Skeleton", side = 3, line = 0.4)
}

draw$diam <- function() {
  terra::plot(panel$root, col = black_ramp, colNA = mask_col, range = c(0, 1),
              legend = FALSE, axes = FALSE, maxcell = Inf, mar = mar_p)
  terra::plot(panel$diam, col = diam_cols, range = diam_rng, legend = FALSE,
              axes = FALSE, maxcell = Inf, add = TRUE)
  draw_inset(full$root, win = wins$diam, col = black_ramp,
             colNA = mask_col, range = c(0, 1))
  draw_inset(full$diam, win = wins$diam, new = FALSE, col = diam_cols,
             range = diam_rng)
  bx <- 0.08 * W; bw <- 0.10 * W; by <- 0.04 * H; bh <- 0.22 * H
  graphics::rect(bx - 0.05 * W, by - 0.02 * H, 0.72 * W, by + bh + 0.05 * H,
                 col = leg_bg, border = NA)
  graphics::rasterImage(grDevices::as.raster(matrix(rev(diam_cols), ncol = 1)),
                        bx, by, bx + bw, by + bh)
  graphics::text(bx + bw + 0.04 * W, by + bh * c(0, 0.5, 1), adj = c(0, 0.5),
                 col = "white", cex = 0.6,
                 labels = format(c(diam_rng[1], mean(diam_rng), diam_rng[2]),
                                 digits = 2))
  graphics::text(bx - 0.03 * W, by + bh + 0.03 * H, "mm", adj = c(0, 0.5),
                 col = "white", cex = 0.65)
  graphics::mtext("Diameter", side = 3, line = 0.4)
}

draw$depth <- function() {
  terra::plot(panel$depth, col = band_cols, colNA = mask_col, legend = FALSE,
              axes = FALSE, maxcell = Inf, mar = mar_p)
  terra::plot(panel$root, col = root_ramp, range = c(0, 1), legend = FALSE,
              axes = FALSE, maxcell = Inf, add = TRUE)
  # Depth changes smoothly, so an inset would show one flat band. The window is
  # marked instead, with the depth range it covers.
  if (zoom_on) {
```

```
    graphics::rect(win[1], win[3], win[2], win[4], border = zoom_col,
                   lwd = zoom_lwd)
    d_rng <- range(terra::values(terra::crop(full$depth, win), mat = FALSE),
                   na.rm = TRUE)
    lab <- sprintf("%.1f–%.1f cm", d_rng[1], d_rng[2])
    tw  <- graphics::strwidth(lab, cex = 0.65)
    th  <- graphics::strheight(lab, cex = 0.65)
    tx  <- min(max(cx, tw / 2 + th), W - tw / 2 - th)
    ty  <- min(win[4] + th, H - 2 * th)
    graphics::rect(tx - tw / 2 - th / 2, ty - th / 2,
                   tx + tw / 2 + th / 2, ty + th * 1.5,
                   col = grDevices::adjustcolor("white", 0.85), border = NA)
    graphics::text(tx, ty, lab, adj = c(0.5, 0), cex = 0.65)
  }
  graphics::legend("bottomleft", bty = "n", fill = band_cols, border = NA,
                   bg = leg_bg, text.col = "white", cex = 0.55, inset = 0.02,
                   legend = paste0(depths, " cm"))
  graphics::mtext("Depth bins", side = 3, line = 0.4)
}

draw$profile <- function() {
  graphics::par(mar = c(4, 4, 1.8, 1))
  plot(depth_data$rootlength.density, depth_data$depth, type = "o", pch = 16,
       ylim = rev(range(depth_data$depth)), col = "#2455A4",
       xlab = expression("cm cm"^-2), ylab = "depth (cm)", cex.axis = 0.8)
  graphics::mtext("Profile", side = 3, line = 0.4)
}
```

A figure rendered from your own scans takes precedence over the bundled example. `overview_figure.R` writes one; put it at `vignettes/figures/overview_panels.png` and the next chunk includes it instead of recomputing. With no such file the bundled scan is used, which is what makes this article reproducible from the package alone.

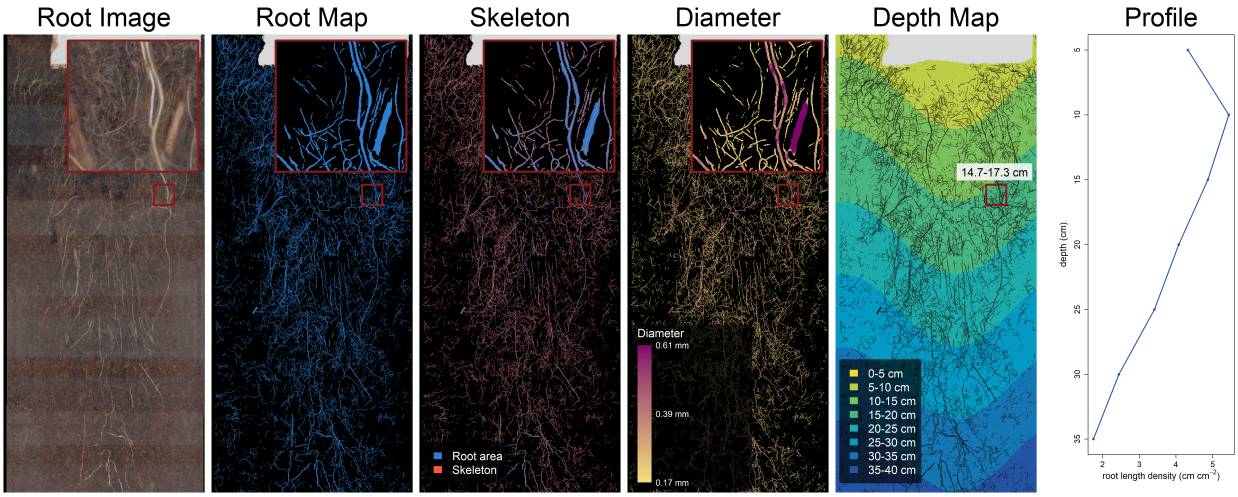


Figure 11: The minirhizotron workflow in six panels.

```
op <- graphics::par(mfrow = c(1, length(panels)), mar = mar_p)
for (pn in panels) draw[[pn]]()
graphics::par(op)
```

The last panel is the point of the other five. Everything to its left is an image; it is the only one with a soil depth axis, and it is what the rest of this article compares between tubes, between sessions and between treatments.

The same profile over a folder

```
result <- root_depth_metrics(
  path_seg            = "scans/segmented/2023_06/",
  path_skl            = "scans/skeleton/2023_06/",
  tube_names          = ids,      # one unique id per file
  dpi                 = 150,
  geometry            = "tube",
  insertion_angles    = 45,       # degrees from vertical
  tube_diameter_cm    = 7,
  soil_starts         = 2.9,      # cm
  depth_interval_cm   = 5,
  bin_round           = "floor",
  session             = "2023_06"
)
```

`path_skl` is optional; left out, each mask is thinned internally, which adds the cost of a skeletonization pass per image and is the single slowest thing this wrapper can be asked to do over a large folder. Supply the segmenter's own skeletons if you have them. Branching order, root angle, color and landscape metrics are separate `calc_*` switches, all off by default because each costs real time too.

Part II: minirhizotron extensions

Everything here still works on a minirhizotron tube, and each section attaches to the path Part I walked: another trait from the same scan, a second session, or the color channels rather than the root shape. The order runs from closest to that path to furthest from it. Part III then changes the imaging setup altogether.

Root angle

Two things have to exist before an angle means anything, and the first is a **direction per root pixel**. It comes from a D8 pass over the depth map restricted to the skeleton: each center-line pixel is assigned whichever of its eight neighbors the depth field falls toward, giving one of eight compass classes. The second is a **reference direction** to score it against, and that is where the assumption sits: the reference is the direction that gets a root away from the tube wall and *downward* fastest, so the whole measure is read from a gravitropic standpoint. A root growing along a nutrient patch or around a stone is not doing worse by this measure in any biological sense; it is simply not doing what gravity would predict.

`deep_drive()` builds both and reports the agreement between them: the fraction of root pixels heading the direction the depth field says is steepest down. It takes an `AngleMap` if you have one from elsewhere, and computes it from `DepthMap` and `RootMap` when you do not, which is the usual case.

```
dd <- deep_drive(DepthMap = depth_map, RootMap = skl)

cat(sprintf("deep_drive: %.3f of root pixels head the deepest way\n", dd))
#> deep_drive: 0.260 of root pixels head the deepest way
```

One number between 0 and 1, so it compares across tubes. It is an agreement score rather than a deviation: a pixel pointing 45 degrees off the steepest direction counts exactly as wrong as one pointing 180 degrees away, since the D8 classes are compared for equality. Two stands could therefore score the same while one grows mostly sideways and the other mostly upward. Weighting each pixel by its angular distance from the reference would separate those, and it would turn the statistic into a mean deviation in degrees; that is not what the function does today, and it is the obvious extension if the distinction matters to you.

It needs a real depth gradient, so it means nothing on a flatbed scan. The batch wrapper reports it as `deep_drive` alongside `mean.steepness.angle`.

`return = "all"` keeps the intermediate rasters. The one worth looking at is the direction assigned to each skeleton pixel, drawn on its own below — the skeleton and the depth bins it was derived from are both shown in Part I.

```
dd_all <- deep_drive(DepthMap = depth_map, RootMap = skl, return = "all")

# deep_drive() returns its map already transposed and flipped: turned the same
# way rot() turns every other panel. So it is used as it comes and NOT rotated
# again, while the root field underneath is the rotated one the workflow figure
# already built. Rotating both is how this panel ends up mirrored.
ang_full <- dd_all$angle_map
terra::ext(ang_full) <- terra::ext(full$root)
ang_full <- terra::ifel(full$skel == 2, ang_full, NA)   # center line only

# Index k + 1 is the class k * 45, so a direction keeps its color from tube to
# tube and two figures compare. 45 and 315 are left out of the wheel: a root
# cannot run up and outward at once, so where they occur at all it is noise.
ang_brk  <- seq(-22.5, 337.5, by = 45)
ang_cols <- c("#FFFFFF", "#FF8A3D", "#FFD60A", "#4D7CFF",
```

```
              "#3DDC84", "#E0218A", "#00E5FF", "#B388FF")
ang_skip <- c(45, 315)

# What is shown. Depth runs down the panel, so the shallow end is the top.
# The first rows are tape and mask rather than soil, and the tube is four
# times longer than it is wide: drawn whole it is a sliver, because terra
# draws at asp = 1 and the page can only give it height. Showing part of the
# length is what buys the scale back.
top_cut   <- 0.05    # drop this much of the length at the shallow end
show_frac <- 0.50    # how much of the length to draw
win_ctr   <- c(0.80, 0.42)   # window center, as fractions of what is shown
win_frac  <- 0.17            # window width, as a fraction of the panel width

y_hi <- H * (1 - top_cut)
y_lo <- max(0, y_hi - show_frac * H)
e    <- terra::ext(0, W, y_lo, y_hi)

hw <- win_frac * W / 2
wx <- min(max(win_ctr[1] * W, hw), W - hw)
wy <- min(max(y_hi - win_ctr[2] * (y_hi - y_lo), y_lo + hw), y_hi - hw)
win <- terra::ext(wx - hw, wx + hw, wy - hw, wy + hw)

a_ang     <- terra::aggregate(ang_full, fact = 4, fun = "modal", na.rm = TRUE)
#> |---------|---------|---------|---------|=========================================
ang_found <- setdiff(sort(unique(terra::values(terra::crop(a_ang, e),
                                               mat = FALSE))), ang_skip)

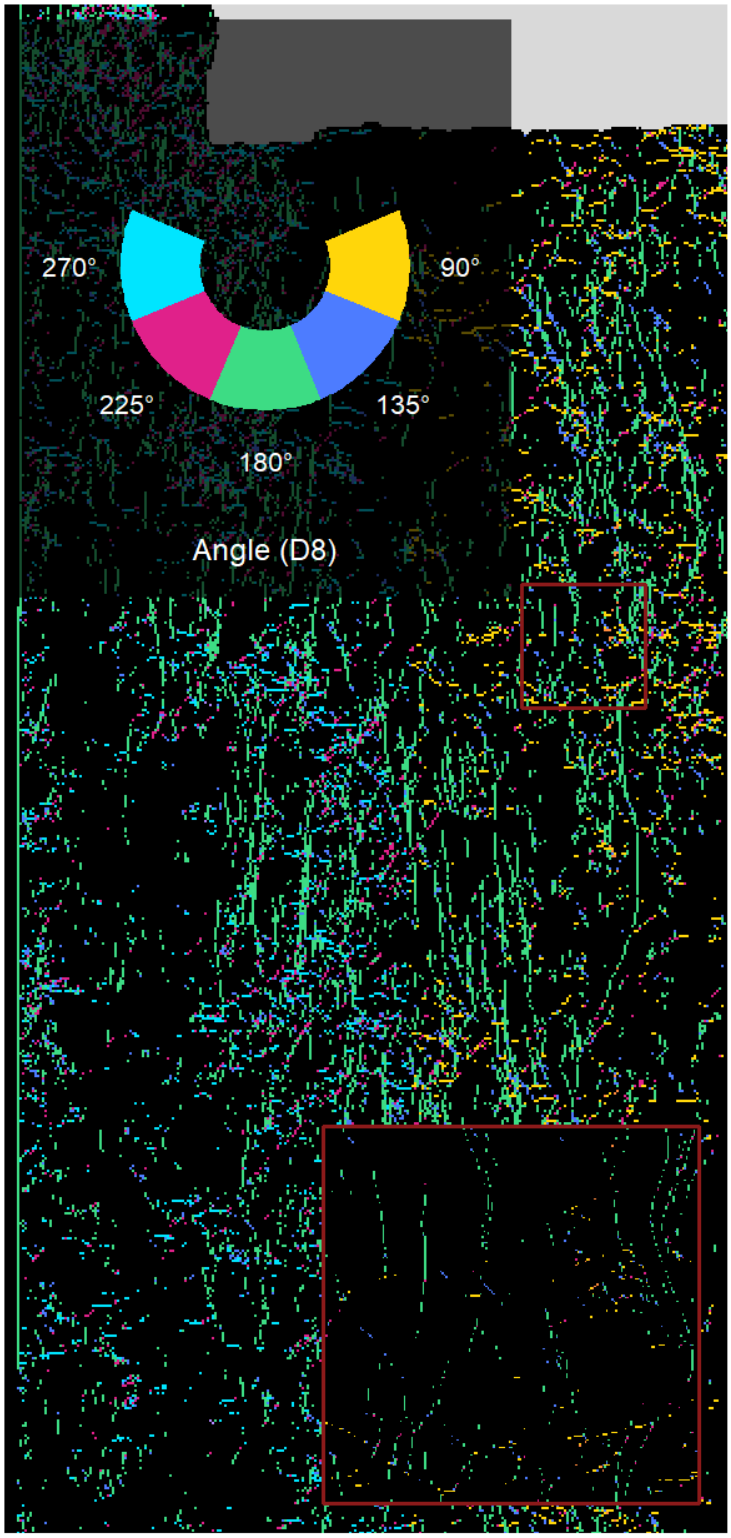
# The same two calls the workflow figure makes: the root raster flat black so
# the field is black and colNA paints only the masked tape, then the
# directions over it as the only color in the panel.
op <- graphics::par(mar = mar_p)
terra::plot(terra::crop(panel$root, e), col = black_ramp, colNA = mask_col,
            range = c(0, 1), legend = FALSE, axes = FALSE, maxcell = Inf,
            mar = mar_p)
terra::plot(terra::crop(a_ang, e), breaks = ang_brk, col = ang_cols,
            legend = FALSE, axes = FALSE, maxcell = Inf, add = TRUE)

# The inset, bottom right of what is shown. inset_e from the helper chunk is
# in whole-frame coordinates, so the box is built against this crop instead.
iw <- 0.52 * W
ie <- terra::ext(e[2] - 0.04 * W - iw, e[2] - 0.04 * W,
                 e[3] + 0.04 * W, e[3] + 0.04 * W + iw)
graphics::rect(ie[1], ie[3], ie[2], ie[4], col = "black", border = NA)
for (lyr in list(list(r = full$root, args = list(col = black_ramp,
                                                 colNA = mask_col,
                                                 range = c(0, 1))),
                 list(r = ang_full,  args = list(breaks = ang_brk,
                                                 col = ang_cols)))) {
  z <- terra::crop(lyr$r, win); terra::ext(z) <- ie
  do.call(terra::plot, c(list(z, add = TRUE, legend = FALSE, axes = FALSE,
                              maxcell = Inf), lyr$args))
}
graphics::rect(ie[1], ie[3], ie[2], ie[4], border = zoom_col, lwd = zoom_lwd)
```

```
graphics::rect(win[1], win[3], win[2], win[4], border = zoom_col,
               lwd = zoom_lwd)

# A wheel rather than a list of numbers: 0 degrees is up and the angle runs
# clockwise, so 180 is straight down the depth axis and 90 and 270 run around
# the circumference. Sectors with no wedge are directions that do not occur.
sz <- 0.20 * W
wc <- c(1.8 * sz, e[4] - 1.8 * sz)
graphics::rect(wc[1] - 1.7 * sz, wc[2] - 2.3 * sz,
               wc[1] + 1.7 * sz, wc[2] + 1.7 * sz, col = leg_bg, border = NA)
for (q in ang_found / 45) {
  tt <- seq((q * 45 - 22.5) * pi / 180, (q * 45 + 22.5) * pi / 180,
            length.out = 24)
  graphics::polygon(wc[1] + c(0.45 * sz * sin(tt), rev(sz * sin(tt))),
                    wc[2] + c(0.45 * sz * cos(tt), rev(sz * cos(tt))),
                    col = ang_cols[q + 1], border = NA)
  graphics::text(wc[1] + 1.35 * sz * sin(q * pi / 4),
                 wc[2] + 1.35 * sz * cos(q * pi / 4),
                 paste0(q * 45, "°"), col = "white", cex = 0.85)
}
graphics::text(wc[1], wc[2] - 1.95 * sz, "Angle (D8)", adj = c(0.5, 0.5),
               col = "white", cex = 0.95)
graphics::mtext("Root direction, shallow end", side = 3, line = 0.4)
```

Root direction, shallow end



```
graphics::par(op)
```

Zero degrees is up and the angle runs clockwise, so 180 is straight down the depth axis and 90 and 270 run around the circumference. A tube at 45 degrees puts most of the deepest-direction mass between 135 and 225, and the fraction agreeing with it is the number `deep_drive()` returns.

Rotation bias and circumferential pattern

Part I censored the rotation. The same slicing answers a second question: are roots spread evenly around the tube, or concentrated on one side? `slice_rotation()` cuts the image into bands of rows in rotation order, and a trait measured per band becomes a sequence indexed by circumferential position.

```
# Trim the shallowest quarter of the depth axis (columns), where the tape and
# the noisiest soil sit. The rotation axis is left whole, since that is what the
# slices below are cut along.
e <- terra::ext(root_layer)
rot_layer <- terra::crop(root_layer,
                         terra::ext(e[1] + e[2] / 4, e[2], e[3], e[4]))

# create circumferential zones
n_slices <- 48
slices <- slice_rotation(rot_layer, n_slices)

terra::plot(rot_layer, maxcell = Inf)

# Every fourth slice boundary, so the bands stay legible: each pair of
# adjacent red lines bounds four of the zones the trait below is measured in.
re    <- terra::ext(rot_layer)
edges <- seq(re[3], re[4], length.out = n_slices + 1)
graphics::abline(h = edges[seq(1, length(edges), by = 4)],
                 col = "red", lwd = 1, lty = 2)
```

```
# tallies from the binary mask: roots = 1, soil/background = 0
n_root <- function(s) {
  v <- terra::values(s, mat = FALSE); sum(v > 0, na.rm = TRUE)
}
n_soil <- function(s) {
  v <- terra::values(s, mat = FALSE); sum(v == 0, na.rm = TRUE)
}

slice_traits <- data.frame(
  slice   = seq_len(n_slices),
  rootlen = sapply(slices, root_length, unit = "cm", dpi = dpi,
                   show_messages = FALSE),  # cm
  rootpx  = sapply(slices, n_root),
  soilpx  = sapply(slices, n_soil)
)

# root length per observed area (root + soil pixels)
slice_traits$rld <- slice_traits$rootlen /
```

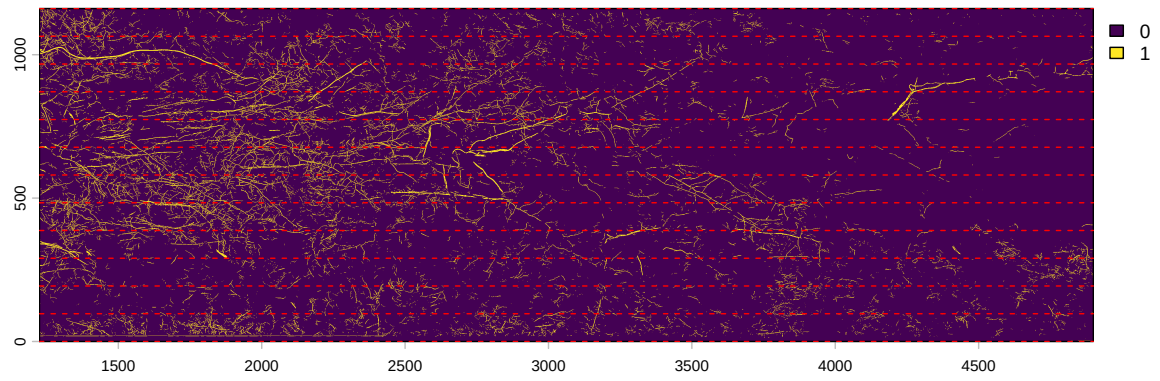


Figure 12: The circumferential slices used below.

```
    ((slice_traits$rootpx + slice_traits$soilpx) *
       (2.54 / dpi)^2)
```

`rhythmicity()` fits `amplitude * sin(2*pi/period * (x + phase)) + offset`, and all three fitted parameters are useful, not just the amplitude:

- **`offset`** is the mean level — the trait averaged around the whole tube, free of the one-sided excess.
- **`amplitude`** is half the peak-to-trough range. The test asks whether it differs from zero, which is the question "is there a side at all?".
- **`phase`** is *where* the peak sits around the circumference. Where the amplitude is clearly non-zero, that is an estimate of the tube's rotational zero from the roots themselves, and it can be compared against the one `estimate_rotation_center()` reads off the tape. Where the amplitude is not significant, the phase is fitting noise and means nothing.

Fixing the period to the number of slices means one full turn of the tube, so a significant amplitude says root abundance varies systematically with circumferential position. Nothing in the fit is specific to time: this is a spatial use of a tool normally introduced for time series, and `res$peakTime` is a slice index rather than a time.

The fit is a Levenberg-Marquardt minimization of the residual function through `minpack.lm::nls.lm`, and the reason is practical rather than principled: base `nls()` takes a formula and routinely fails with a singular gradient on a sine with three free parameters and ordinary starting values. `nls.lm` minimizes the residuals directly and converges where `nls()` gives up. The amplitude tests follow the likelihood-based rhythmicity tests of diffCircadian (Ding *et al.* 2021); `method = "F"`, `"LR"` and `"Wald"` are the three that package distinguishes.

```
res <- rhythmicity(
  x          = slice_traits$slice,
  y          = slice_traits$rld,
  fix_period = n_slices,
```

```
  method     = "F"
)

cat(sprintf("Mean %.4f | Amplitude %.4f | Phase %.2f | R2 %.3f | p %.4f\n",
            res$offset, res$amplitude, res$phase, res$R2, res$p_value))
#> Mean 4.9233 | Amplitude 1.8943 | Phase 39.56 | R2 0.650 | p 0.0000
```

```
# Smooth fitted sine over the slice index
fit_df <- data.frame(slice = seq(1, n_slices, length.out = 200))
fit_df$rld <- res$amplitude *
              sin(2 * pi / n_slices * (fit_df$slice + res$phase)) +
              res$offset

ggplot(slice_traits, aes(slice, rld)) +
  geom_point(color = "gray50") +
  geom_line(data = fit_df, color = "coral", linewidth = 1) +
  geom_hline(yintercept = res$offset, color = "gray40", linetype = 3) +
  annotate("text", x = 1, y = res$offset, hjust = 0, vjust = -0.5,
           size = 3, color = "gray30", label = "offset (mean around the tube)") +
  annotate("segment", x = res$peakTime, xend = res$peakTime,
           y = res$offset, yend = res$offset + res$amplitude,
           linetype = 2, color = "firebrick3") +
  annotate("text", x = res$peakTime, y = res$offset + res$amplitude,
           hjust = -0.08, vjust = 0.5, size = 3, color = "firebrick3",
           label = sprintf("peak at slice %.1f\n= fitted top of the tube",
                           res$peakTime)) +
  annotate("text", x = Inf, y = Inf, hjust = 1.05, vjust = 1.5,
           label = sprintf("A = %.4f, R2 = %.3f, p = %.4f",
                           res$amplitude, res$R2, res$p_value)) +
  theme_minimal() +
  labs(x = "Circumferential slice (1 = top, going around the tube)",
       y = "Root Length Density (cm / cm2)",
       title = "Root distribution around tube circumference")
```

```
# cross section view
inner <- 6
ggplot(slice_traits, aes(slice, rld + inner)) +
  geom_col(fill = "steelblue") +
  # In polar coordinates a vertical line is a radius, so this marks the fitted
  # peak: where the sine says the top of the tube is.
  geom_vline(xintercept = res$peakTime, color = "firebrick3",
             linetype = 2, linewidth = 0.8) +
  coord_polar(start = pi) +
  xlab("") + ylab("") +
  ggtitle("Circumferential Root Length Distribution") +
  theme_light() +
  theme(axis.text = element_blank()) +
  annotate("rect", xmin = -Inf, xmax = Inf, ymin = 0, ymax = inner,
           fill = "white", color = NA)
```

A significant, non-trivial amplitude means more root material on one side of the tube than the other. Gravitropism and the installation angle are the biological and geometric causes, but check the methodological

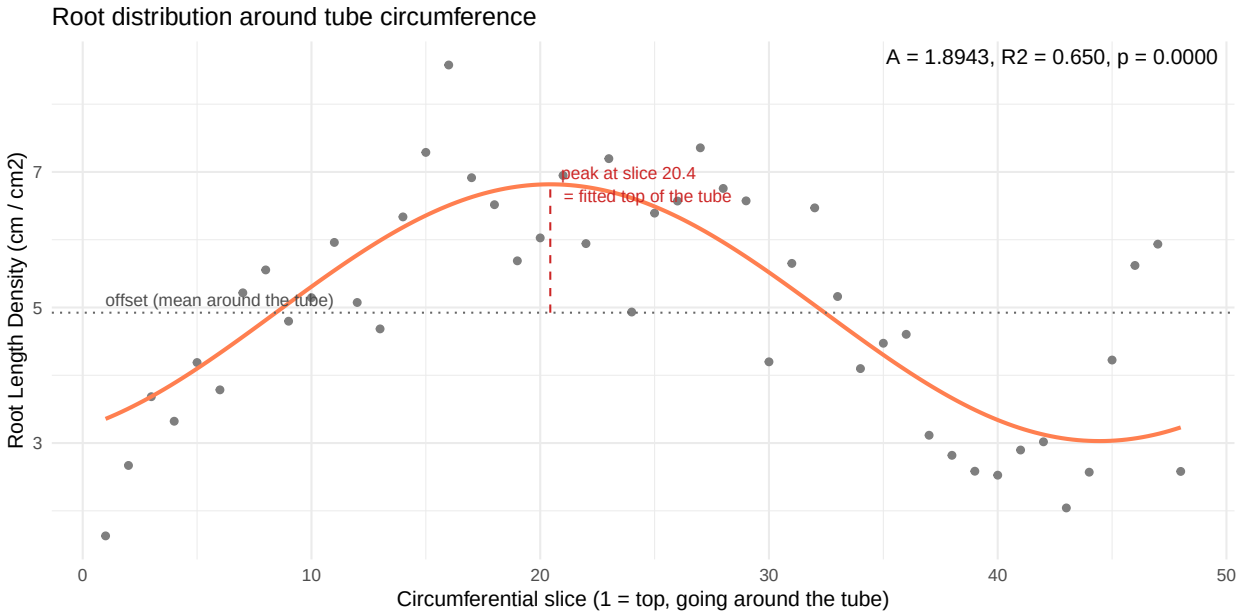


Figure 13: Root length density around the circumference, with the fitted sine.

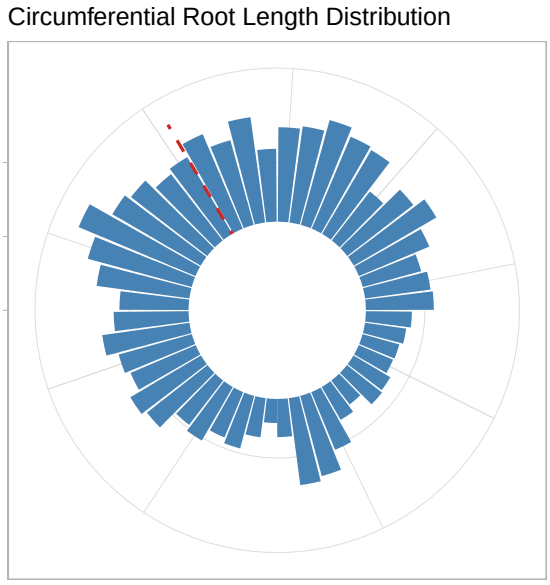


Figure 14: The same distribution on polar axes.

one first: tape on the upper side of the tube, inside the soil, hides whatever grew under it. That removes root from one band of rows and from the shallowest depths at the same time, which produces exactly the signal a real one-sided bias produces. Comparing the fitted `phase` against the rotational zero from `estimate_rotation_center()` is the check — a peak sitting where the tape is is an artifact, one sitting elsewhere is not.

Comparing depth profiles

The usual way to say two stands root differently is to compare mean rooting depth. `MRD()` gives one number per profile, it is easy to model, and it is the right summary when the thing that changed is how deep the root sits. Its limit is that it is a center of mass: it collapses a whole profile to one coordinate, so any two profiles with the same center are the same number to it. A stand with all its root at 20 cm and a stand with half at the surface and half at 40 cm have the same mean rooting depth and nothing else in common.

Comparing the distributions keeps what the mean throws away. The earth-mover distance asks how far the root mass would have to be carried, down the depth axis, to turn one profile into the other: **distance here means depth**, in centimeters. A profile shifted 35 cm costs more than one shifted 10 cm, and a profile of a different shape costs whatever rearranging it takes. It reduces to the difference in mean rooting depth exactly when all the root moves the same way, and exceeds it whenever some moves up and some down — so it is never less informative than `MRD()`, and the gap between the two is itself the quantity that says the shapes differ.

In one dimension that has an exact picture. Plot each profile as the cumulative share of total root length from the surface down, and the area between the two curves *is* that distance. At any depth the horizontal gap between the curves is the share of root that has to cross that depth; adding it up down the profile gives the total, in centimeters.

```
z  <- seq(0, 60, by = 5)                    # 13 intervals, 5 cm each

# A surface-weighted reference, with no mass below 25 cm. Keeping the support
# in the upper half of the grid matters: a profile shifted past the bottom of
# the grid loses the mass that falls off, and renormalizing what is left
# changes its shape, so it is no longer the same profile moved down.
p0 <- exp(-abs(z - 10) / 7)
p0[z > 25] <- 0
reference <- p0 / sum(p0)

# Move the whole profile down by k intervals.
shift_by <- function(p, k) c(rep(0, k), head(p, length(p) - k))

shifted10 <- shift_by(reference, 2)         # 2 intervals = 10 cm
shifted35 <- shift_by(reference, 7)         # 7 intervals = 35 cm
even      <- rep(1 / length(z), length(z))  # equal root in every interval

# A two-layer profile: a surface mode and a second one at 45 cm. The weight on
# the deep mode is solved so this profile has exactly the reference's mean
# rooting depth -- same center of mass, different shape.
shallow <- exp(-z / 5);           shallow[z > 15] <- 0
deep    <- exp(-abs(z - 45) / 7); deep[z < 30]    <- 0
shallow <- shallow / sum(shallow)
deep    <- deep / sum(deep)

mean_depth <- function(p) sum(z * p)
w <- (mean_depth(reference) - mean_depth(shallow)) /
```

```
     (mean_depth(deep)      - mean_depth(shallow))
bimodal <- (1 - w) * shallow + w * deep

# The package function, with the depth of each bin as `index` and no tail
# weighting, so every centimeter of depth counts the same. The result is in the
# units of `index`, which is centimeters here.
emd_cm <- function(a, b)
  compare_depth_distribution(a, b, metric = "wasserstein", index = z,
                             tail_weight = FALSE)

compare_panel <- function(b, label_b, title) {
  Fa <- cumsum(reference)
  Fb <- cumsum(b)
  plot(NA, xlim = c(0, 1), ylim = rev(range(z)),
       xlab = "cumulative share", ylab = "depth (cm)", main = title)
  graphics::polygon(c(Fa, rev(Fb)), c(z, rev(z)), col = "grey88",
                    border = NA)
  graphics::lines(Fa, z, col = "steelblue4", lwd = 2)
  graphics::lines(Fb, z, col = "coral3",     lwd = 2)
  graphics::legend("bottomright", bty = "n", lwd = 2, cex = 0.8,
                   col = c("steelblue4", "coral3"),
                   legend = c("reference", label_b))
  graphics::mtext(sprintf("%.1f cm to move", emd_cm(reference, b)),
                  side = 3, line = -1.3, cex = 0.75)
}

op <- graphics::par(mfrow = c(2, 2), mar = c(4, 4, 3, 1))
compare_panel(shifted10, "10 cm deeper",      "same shape, 10 cm down")
compare_panel(shifted35, "35 cm deeper",      "same shape, 35 cm down")
compare_panel(even, "equal at all depths",    "spread evenly")
compare_panel(bimodal, "bimodal",            "same mean depth, other shape")
```

```
graphics::par(op)
```

The first two panels hold the same root in the same shape and differ only in how far down it sits, so the band widens in proportion – and they are the check on the metric: a profile moved down by a fixed distance has to return that distance, and 10 cm and 35 cm are what the shaded areas measure. The third holds the same amount of root in every interval instead of peaking near the surface, so its cumulative curve is a straight line: the band is a different shape rather than simply wider, because the mass has to be spread out as well as moved. There is no shift to report for it, and its 19.3 cm is what rearranging one profile into the other costs.

The fourth panel is the one to read twice. It holds a bimodal profile – a surface mode and a second one at 45 cm – with the weight on the deep mode solved so that its mean rooting depth is exactly the reference's.

```
c(reference = mean_depth(reference), bimodal = mean_depth(bimodal))
#> reference   bimodal
#>  10.68323  10.68323
emd_cm(reference, bimodal)
#> [1] 9.669479
```

`MRD()` reports no difference at all between these two stands. The distance is 9.7 cm, and every centimeter of it is shape.

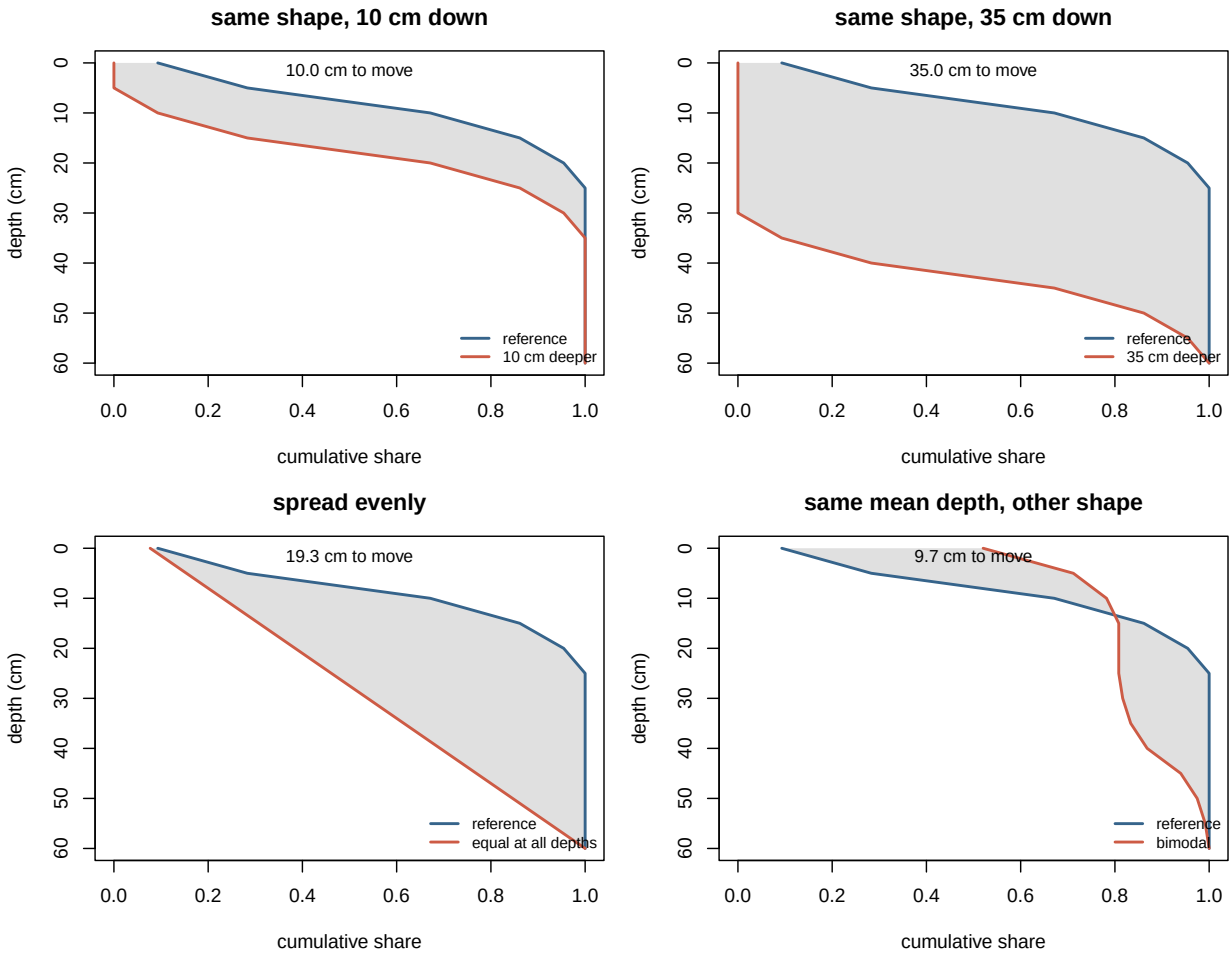


Figure 15: Transport distance as the area between cumulative profiles.

The curves show why. In the first three panels they never cross: one profile is deeper than the other at every depth, all the root travels the same way, and the distance comes out equal to the difference in mean rooting depth – 10, 35 and 19.3 cm. In the fourth they cross, because the bimodal profile holds more root than the reference near the surface and more again at depth. Mass has to move up as well as down; the two directions cancel in the mean and add in the distance. Crossing cumulative curves are the visual test for a difference that mean rooting depth cannot see.

`tail_weight = TRUE` is the reason the function takes more than two vectors. It multiplies the cost of moving mass by a weight that varies with depth, so a centimeter of displacement deep in the profile can be made to count for more than a centimeter near the surface — `weighting` picks the shape and `parameter` its steepness. Left at `FALSE`, as above, every depth counts equally and the result is the plain earth-mover distance the figure draws.

`compare_depth_distribution()` is the function for this, with `metric = "wasserstein"` (default), `"js"` or `"kl"`, and optional tail weighting so deep layers can count for more than shallow ones. Jensen-Shannon and Kullback-Leibler compare bin by bin and have no notion that one bin is adjacent to another, so two profiles that do not overlap score much the same whether they are 10 cm or 50 cm apart.

The shaded areas above and the function agree: with `index` giving the depth of each bin, `compare_depth_distribution()` returns the mass-weighted mean distance the root has to travel, so a profile shifted 10 cm returns 10. Interpreting it needs two things kept in mind. The profiles are normalized, so total root amount divides out – a tube with ten times the root but the same depth distribution scores 0, and abundance has to be reported separately. And the value is bounded below by the difference in mean rooting depth: two tubes 3 cm apart in `MRD()` are at least 3 cm apart here. That bound is tight whenever the cumulative curves do not cross, which is the ordinary case for a profile that simply deepened. What is informative is the gap: a distance well above the difference in mean depth means root moved both up and down, so the shapes differ and not just the center of mass. The two-layer panel above is that case taken to its limit – the difference in mean depth is zero and the whole distance is shape.

Turnover between sessions

Turnover is how much of the root system is replaced over time: root produced and root lost between two visits to the same tube, rather than how much is there. It matters because standing root length can sit still while the roots making it up are entirely different ones, and that difference is what carbon and nutrient budgets turn on.

`root_turnover()` offers two methods, for two kinds of input, and they do not measure the same thing.

`method = "tc"` — *temporal comparison.* Two segmented images of the same tube from consecutive sessions. It measures root amount in each and takes the difference, so what you get is **net** change.

`method = "dpc"` — *decay, production, constant.* One image in which a segmenter has already matched roots between the two sessions and labeled every root pixel as decayed, newly produced, or unchanged. Because it separates the two directions, what you get is **gross** change.

Use `"tc"` when all you have is a pair of segmentations, which is the common case, and read the result as a lower bound. Use `"dpc"` when your segmenter produces the matched coding, and read it as an upper bound. The reason for both caveats is the same: `"tc"` cannot see root that appeared and vanished between two scans and so **understates** turnover, while `"dpc"` counts every labeled difference including the ones that come from imperfect matching and segmentation noise, and so **overstates** it. The truth is between them, and reporting which method produced a number matters more than the number.

```
data(skl_Oulanka2023_Session01_T067)
t1 <- load_flexible_image(skl_Oulanka2023_Session01_T067,
                          output_format = "spatrast")
t2 <- load_flexible_image(skl_Oulanka2023_Session03_T067,
                          output_format = "spatrast")
```

```
root_turnover(img1 = t1, img2 = t2, method = "tc", tc_method = "kimura",
              dpi = dpi, unit = "cm")
#>   standingroot_t1 standingroot_t2 production newroot.per_t1 newroot.per_t2
#> 1        17875.84        16835.23  -1040.613        -0.0582        -0.0618
```

`tc_method` picks how "root amount" is measured — `"kimura"` for length, pixel counts for area — and the two can disagree in sign when roots thicken without extending.

```
data(TurnoverDPC_data)
dpc <- load_flexible_image(TurnoverDPC_data, output_format = "spatrast",
                           scale = "none")
root_turnover(dpc, method = "dpc")
#>     tape constant production   decay newgrowth.ratio decay.ratio constant.ratio
#> 1 887012   438681    2770138 3355722          0.8633      0.8844         0.0668
```

`include_virtualroots` switches the denominator between "present at this time point" and "present at any time point". The second includes root that was there only in the other session, which lowers every ratio; the choice changes the numbers and has to be reported with them.

Soil color, texture, and the rhizosphere

Everything that reads the color channels rather than the root shape. `classify_soil_rgb()` assigns every pixel to a material class by nearest-centroid matching in CIE LAB, leaving pixels too far from any centroid unclassified rather than forcing them into the nearest class.

```
# downsample_fact speeds up the demo; drop it for full resolution
result <- classify_soil_rgb(rgb, downsample_fact = 4, verbose = FALSE)

# Raw beside classified over one window: a class map is only checkable against
# the colors it was derived from.
op <- graphics::par(mfrow = c(1, 2), mar = mar_p)
zoom_plot(rgb,        main = "Scan",       frac = 0.25, center = c(0.6, 0.45),
          overview = FALSE)
zoom_plot(result$map, main = "Classified", frac = 0.25, center = c(0.6, 0.45),
          overview = FALSE)
```

```
graphics::par(op)
```

```
result$metrics
#>           class n_pixels pct_pixels L_mean A_mean B_mean L_sd A_sd B_sd R_mean G_mean B_mean_rgb
#> 1     dark_soil    40252     11.331    8.8    1.5    2.4 3.38 1.45 1.35   28.3   23.9       20.9
#> 2      red_soil   233641     65.768   13.9    4.4    5.9 3.55 1.99 2.10   44.2   33.0       27.1
#> 3          root     5707      1.606   34.5    4.6   10.7 8.19 3.26 5.74   94.7   78.4       64.6
#> 4   silver_tape    63821     17.965   73.1   -2.0    0.9 3.63 0.55 1.06  176.4  180.6      177.8
#> 5 coarse_debris    11632      3.274   22.5    7.9   11.9 3.57 2.96 3.39   70.4   49.1       36.5
#> 6  unclassified      197      0.055   20.1   18.6   22.7 7.00 2.11 4.84   80.2   36.5       13.9
#>   mean_dist_to_centroid hex_actual
#> 1                  5.51    #1C1816
#> 2                  4.32    #2C211B
#> 3                 10.11    #5F4E40
```

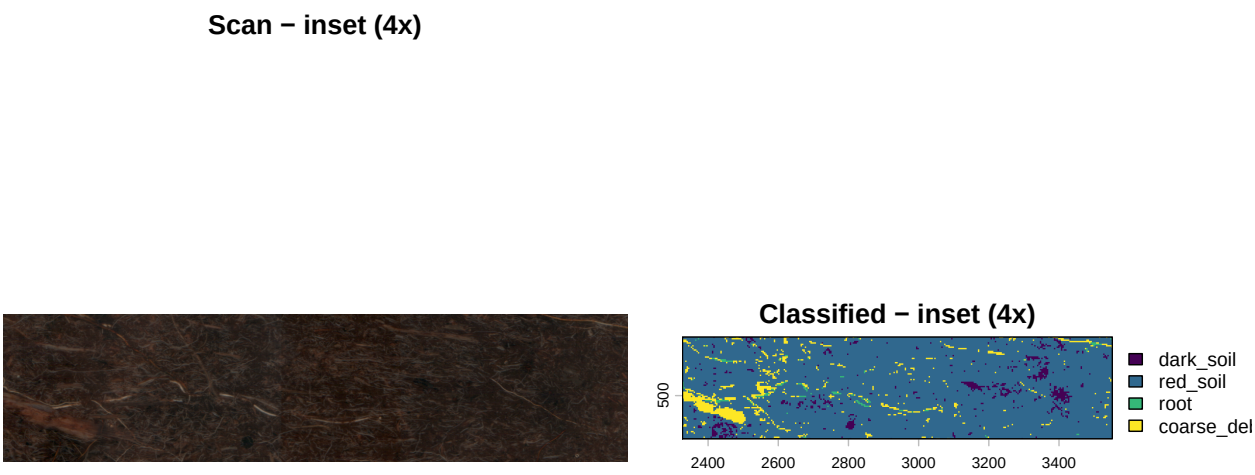


Figure 16: Scan and class map over the same window.

```
#> 4                  7.67    #B0B5B2
#> 5                  6.38    #473124
#> 6                    NA    #50240F
```

```
plot_soil_classification(result)
```

The returned list carries per-class statistics: pixel counts, area fractions, mean LAB and RGB colors, and the mean distance to the centroid. Watch the unclassified fraction – a high value means the shipped centroids do not match your scanner, and `build_soil_centroids()` derives your own from color picks. `plot_soil_classification()` is the figure to read it from: the class map, a legend in dendrogram order, and a dendrogram of inter-class LAB distances, so you can see which classes sit close enough together to be confused. `color_mode` switches the palette: `"contrast"` is the default and the one used here, `"vibrant"` saturates it, and `"centroid"` paints each class in the mean color of the pixels assigned to it — computed from the image, not chosen — which looks faithful but puts several soil classes in nearly the same red-brown, so it is the hardest of the three to read.

Texture is delegated to **glcm**: `analyze_soil_texture()` hands it the band and returns what it computes. `grays` sets how many grey levels the image is reduced to and `window` the size of the moving window; both are glcm's own arguments, passed straight through.

```
tex <- analyze_soil_texture(
  rgb,
  grays   = 12,
  window  = c(5, 5),
  metrics = c("variance", "second_moment", "correlation", "entropy")
```

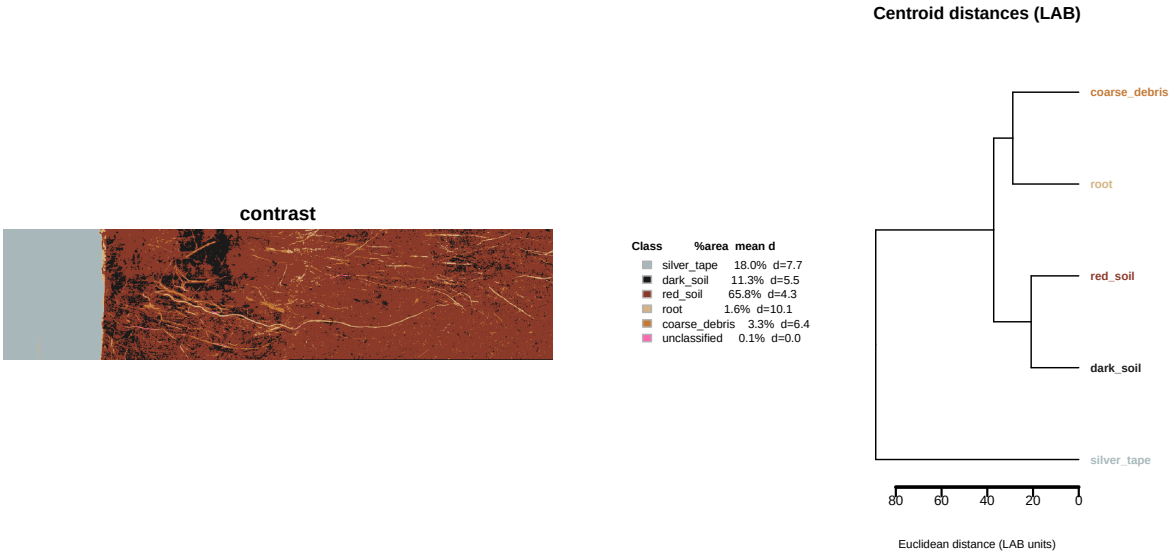


Figure 17: Soil classes with their distances in LAB color space.

```
)
terra::plot(tex)
```

```
halo <- create_root_buffer(root_layer, width = 3, halo_only = TRUE,
                           kernel = "circle")
zoom_plot(halo, main = "Rhizosphere halo (width 3)", frac = 0.1)
```

```
halo10 <- create_root_buffer(root_layer, width = 10, halo_only = FALSE,
                             kernel = "circle")
zoom_plot(halo10, main = "Roots + halo (width 10)", frac = 0.25,
          overview = FALSE)
```

`kernel` switches between a `"circle"` (8-neighbor) and a `"diamond"` (4-neighbor) growth shape, and `width` is the number of dilation iterations.

```
tube_coloration(rgb)
#>      rcc    gcc   bcc        hue saturation luminosity      red    green    blue
#> 1 0.4117 0.3192 0.269 0.06635659  0.1982295    61.0889 67.84946 59.75458 54.3997
```

With roots, their immediate surroundings and the rest of the frame separated by the halo above, the same color measurement can be taken three times and compared.

```
# A 10 px halo with roots excluded, so the three groups do not overlap.
halo_only <- create_root_buffer(root_layer, width = 10, halo_only = TRUE,
                                kernel = "circle")

# The unweighted mean of the three channels. This is not luminosity: a
```

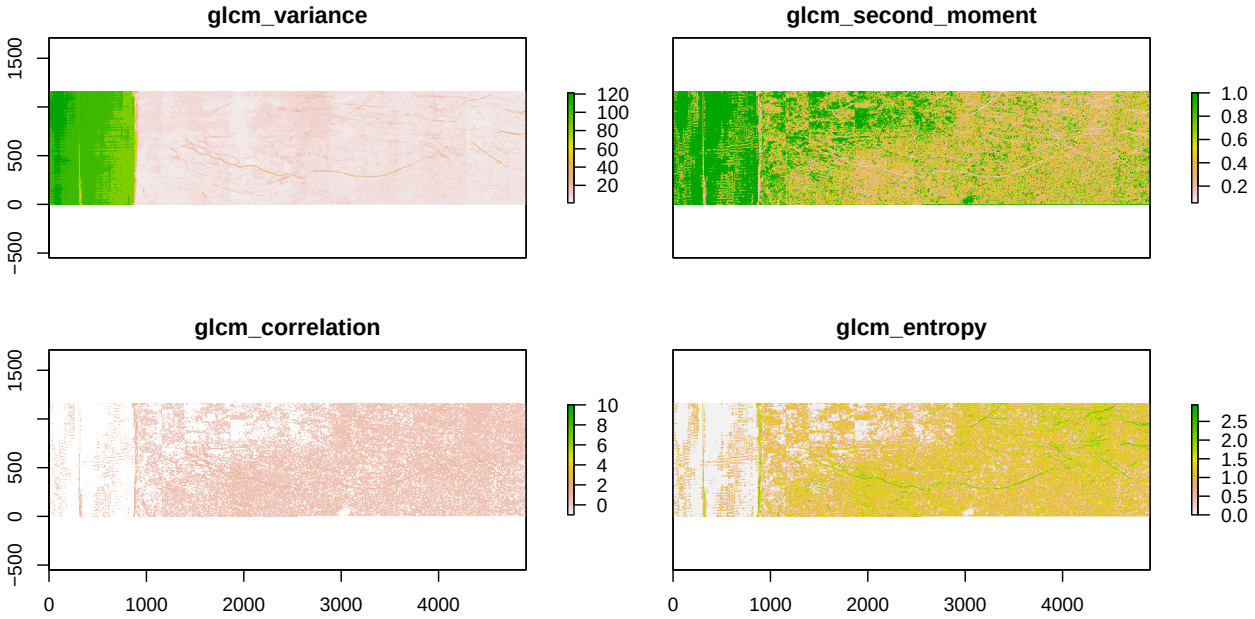


Figure 18: Four GLCM texture rasters of the scan.

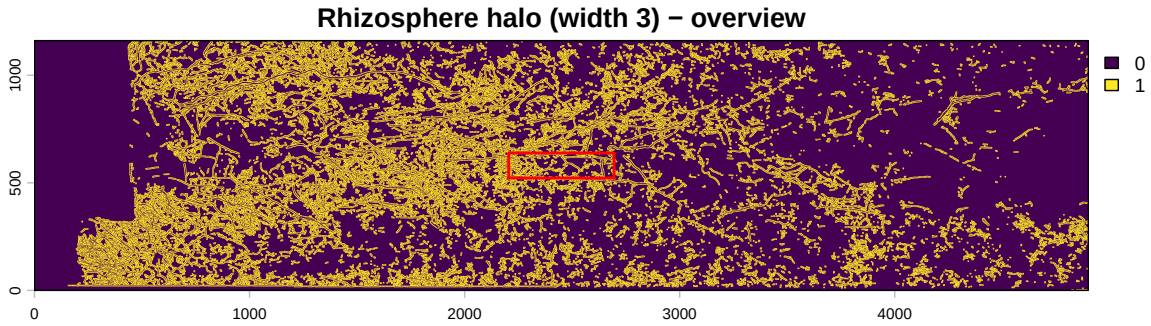


Figure 19: A rhizosphere halo three pixels wide.



Figure 20: The same halo at native resolution.

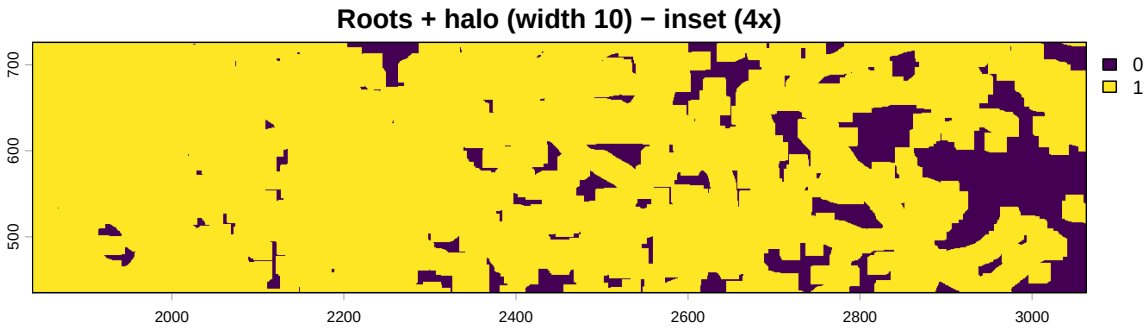


Figure 21: Roots with a halo ten pixels wide.

```r
# perceptual luminance weights green far above blue, and `tube_coloration()`
# reports that weighted version. The unweighted mean is used here because the
# question is which channels differ, and a weighting would hide it.
bright <- terra::app(rgb_disp[[1:3]], fun = mean, na.rm = TRUE)
b <- terra::values(bright,     mat = FALSE)
r <- terra::values(root_layer, mat = FALSE) == 1
h <- terra::values(halo_only,  mat = FALSE) == 1
r[is.na(r)] <- FALSE
h[is.na(h)] <- FALSE

# Take the values, then drop NA, then thin: subsetting first means a group can
# never hand density() a missing value, and sampling values rather than indices
# avoids sample()'s behavior on a length-one vector.
set.seed(1)
pick <- function(sel, n = 40000) {
  v <- b[sel]
  v <- v[!is.na(v)]
  if (length(v) > n) sample(v, n) else v
}

groups <- list(root = pick(r), halo = pick(h), peat = pick(!r & !h))
dens   <- lapply(groups[vapply(groups, length, 1L) > 1L], stats::density)

plot(NA, xlim = range(unlist(lapply(dens, function(d) d$x))),
     ylim = c(0, max(vapply(dens, function(d) max(d$y), 1))),
     xlab = "brightness (mean of R, G, B)", ylab = "density", main = "")
cols <- c(root = "#C1440E", halo = "#2455A4", peat = "#8C6D46")
for (nm in names(dens)) graphics::lines(dens[[nm]], col = cols[[nm]], lwd = 2)
graphics::legend("topright", bty = "n", lwd = 2, col = cols[names(dens)],
                 legend = c(root = "root", halo = "halo around roots",
                            peat = "bulk peat")[names(dens)])
```

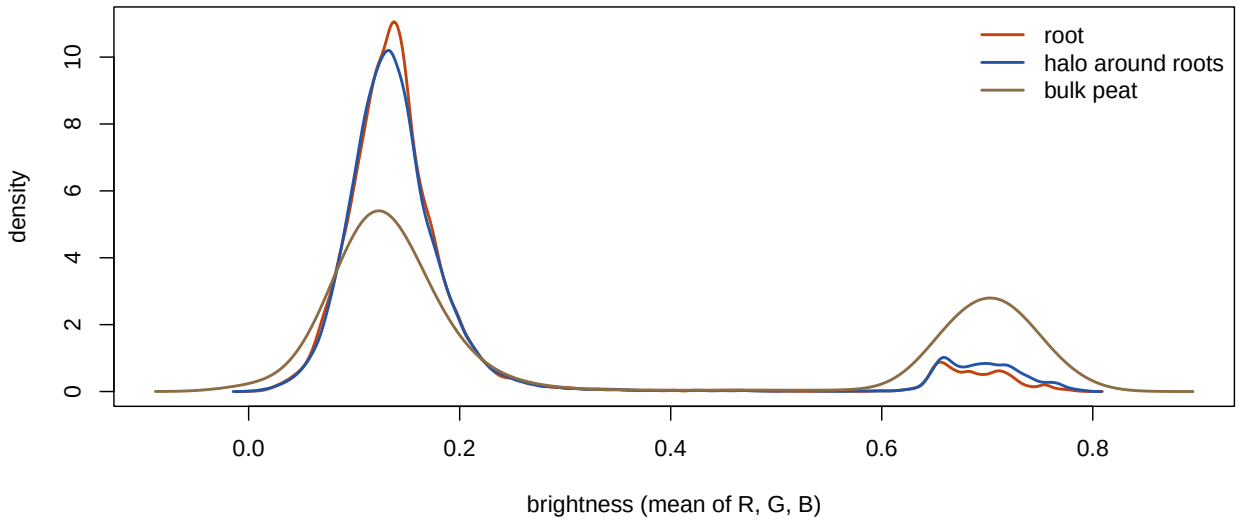


Figure 22: Brightness of root, halo and background pixels.

Three groups, one measurement. Whether the halo sits between root and bulk peat, or on top of one of them,

is the thing to look at: a halo that tracks bulk peat says the roots are not visibly altering their surroundings at this resolution, and a halo displaced toward the root curve says they are — or that the segmentation is a pixel or two generous and the halo is partly root. Widening `width` and watching whether the curve migrates separates those two readings.

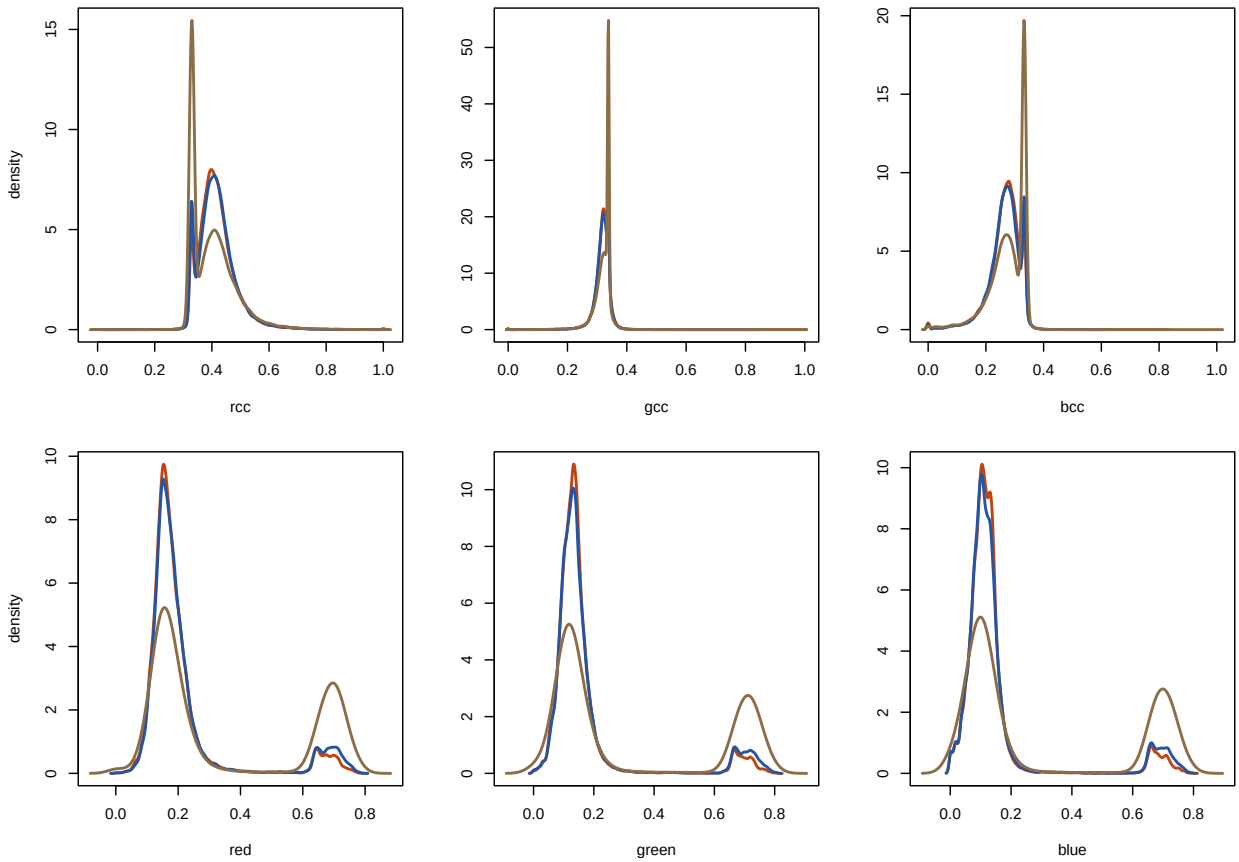
The bulk-peat curve is bimodal, and that is the scan rather than the soil. A stitched mosaic is built from frames exposed separately, and the brightness bands running across the scan in the first figure of Part I are those frames meeting. Each exposure contributes its own mode. It matters here because any color comparison between depths is partly a comparison between frames, so a depth trend in color should be checked against where the joins fall before it is read as soil chemistry. Per channel the same split shows up:

```
# Chromatic coordinates, the same quantities tube_coloration() reports: each
# channel over the sum of the three. Dividing out the total removes exposure,
# which is what made the brightness curve bimodal, so what is left is color.
tot <- terra::app(rgb_disp[[1:3]], fun = sum, na.rm = TRUE)
cc  <- lapply(1:3, function(k) terra::values(rgb_disp[[k]] / tot, mat = FALSE))
names(cc) <- c("rcc", "gcc", "bcc")

# Raw channels underneath, for comparison: same roots, same halo, same peat.
raw <- lapply(1:3, function(k) terra::values(rgb_disp[[k]], mat = FALSE))
names(raw) <- c("red", "green", "blue")

curves <- function(vals) lapply(list(root = r, halo = h, peat = !r & !h),
  function(sel) {
    v <- vals[sel]; v <- v[is.finite(v)]
    stats::density(if (length(v) > 40000) sample(v, 40000) else v)
  })

op <- graphics::par(mfrow = c(2, 3), mar = c(4, 4, 2, 1))
for (k in c(names(cc), names(raw))) {
  dk <- curves(if (k %in% names(cc)) cc[[k]] else raw[[k]])
  plot(NA, xlim = range(unlist(lapply(dk, function(d) d$x))),
       ylim = c(0, max(vapply(dk, function(d) max(d$y), 1))),
       xlab = k, ylab = if (k %in% c("rcc", "red")) "density" else "",
       main = "")
  for (nm in names(dk)) graphics::lines(dk[[nm]], col = cols[[nm]], lwd = 2)
}
```




```
graphics::par(op)
```

Because each coordinate is a share of the total, the frame-to-frame exposure differences divide out and what separates the groups is color rather than lighting. A group that shifts in `rcc` but not in `gcc` or `bcc` has changed hue; one that does not shift in any of the three differs only in brightness, which the previous figure already showed.

`create_root_buffer()` grows a buffer around every root pixel, a simple proxy for the rhizosphere. With `halo_only = TRUE` it returns the ring alone.

Part III: flatbed scans

Everything to this point assumed a tube. A flatbed scan of washed roots is the other half of the pair described in the introduction: no depth axis, and architecture in its place.

The same traits, without a depth axis

A flatbed scan of washed roots is the other geometry the package serves, and it is a special case rather than a different pipeline: the same cleaning, skeleton, length and diameter functions apply, and only the depth axis goes away. Where a root lies on the tray says nothing about where it grew, so the whole scan is one observation.

The raw segmentation below is worth a moment before it is cleaned. It carries two kinds of flaw that `clean_image()` addresses and that are easy to miss at full-scan scale: isolated specks of a few pixels, which the tracer would read as one-segment root systems, and pinholes inside roots, which survive skeletonization as small loops and show up later as `artefact_cycles`. The magnified pair shows both.

What that buys is architecture. A washed root system is rarely complete — fragments break off at the base and at branch points during excavation and washing, so how much you recover depends on the setup. What you do have is a set of individually complete, fully connected units laid flat and unoccluded, and that is enough for a branching graph to mean something. On a tube it is not: roots enter and leave the field of view, so the graph describes visible fragments rather than root systems.

Junctions and crossings

Two situations look identical in a binary image and are biologically opposite. A **junction** is a root branching: three arms meet, one parent and two children. A **crossing** is two roots lying over one another: four arms meet, nothing branched, and the two roots merely touch in projection. Counted naively, a crossing contributes a branch point that does not exist, two tips that do not exist, and a closed loop that does not exist.

`root_graph_pipeline()` handles both, and prints what it did: tracing finds one segment per arm, `spliced pass-through nodes` merges arms that are not really ends, and `resolved crossings` pairs up the arms of a crossing by direction so each root continues *through* the node. `network_metrics()` then reports whatever could not be paired as `artefact_cycles` — loops in a structure that cannot have them, which is a measurement of unresolved overlap rather than a property of the plant.

Which rule applies at a node is decided first by **how many arms meet there**. From four arms up, the node can be joined into roots passing through it. At three arms nothing is ever joined; what remains there is the question of which arm continues the root and which arm is a branch off it, which `assign_root_order()` decides from straightness and diameter together, weighted by `diam_weight`.

The crossing rules, in plain terms:

1. **Four arms or more.** `resolve_crossings()` considers any node where at least four arms meet. A three-arm node is a junction to begin with and is left to the ordering rule.
2. **Each arm gets a direction**, read `look = 5` pixels back along it and pointing away from the node.
3. **A pair of arms is scored by how nearly opposite its two directions run.** The score is a dot product: -1 means the pair runs straight through the node, 0 means a right-angle turn. A pair may be joined when it scores below `crossing_straight` (default -0.5, so a root may bend by at most 60 degrees across the node).
4. **As many arms are paired as can be.** Of the ways to pair up the arms, the one pairing the most arms wins, ties going to the straightest total. Four arms give two pairs or one; six arms meeting can be three roots crossing.

5. **An arm left over is a lateral**, and the pair it leaves is held back from splicing, so the lateral stays attached and the node is read as an ordinary junction. The pair holding the arm nearest in direction is the one held, the thinner pair breaking a tie.
6. **`crossing_diam_ratio` can also require the paired roots to be comparably thick**, which separates a crossing from a branch point whose two laterals happen to leave on opposite sides. It is `0`, off, by default.

Note what rule 4 does *not* read: the angle **between** the two roots. A square crossing and a 20-degree one are the same case, because in each the paired arms run opposite to one another. What is read is how straight each root stays across the node.

What that does to real nodes

`data-raw/node-gallery.R` is a standalone script, needing no scans, that draws ten nodes spanning three to six arms and runs each through the pipeline at its default settings. Each hue is one root; a lateral is a **darker tone of the root it leaves**, and an axis is drawn wider than its branches. Under each panel is what came back.

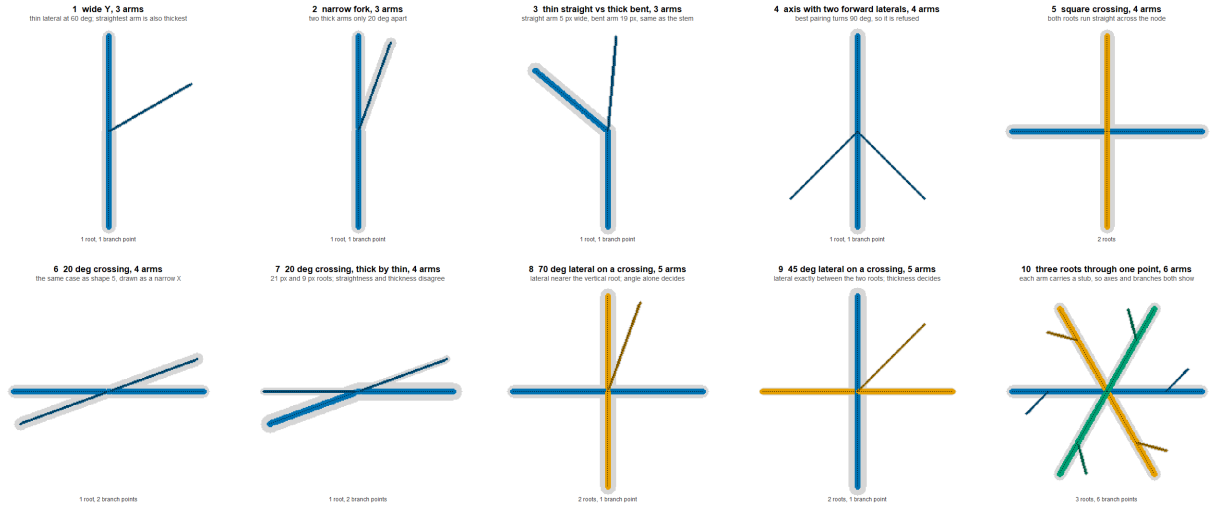


Figure 23: Ten synthetic nodes as the pipeline reads them.

Counting follows the node, not the pixel. **Every lateral arm leaving a node is one branch point**, so two laterals at one pixel are two branch points, and **every pair of roots that overlap at a node is one junction point**, so three roots through one pixel are three junctions. A node can hold both at once, and neither count swallows the other.

Each node is settled by two numbers: the score of the arrangement the rule picks and the score of the runner-up. A wide gap means the reading is firm, a narrow one that the image barely separates two answers. The script prints them all; the four-arm nodes read:

node	chosen	runner-up	gap
square crossing	−2.00	0.00	2.00
axis with two forward laterals	−1.00	0.00	1.00
crossing at 20 degrees	−2.00	−1.88	0.12
the same, 21 px by 9 px	−2.00	−1.88	0.12

The axis with two forward laterals is the one refused: its total is the best available, but acceptance tests the *worst single pair*, and that pair turns a right angle at 0.00, well above −0.5. The two 20-degree crossings are accepted on a gap of 0.12 out of 2, and they come out identically although their widths differ — the pairing reads directions only, so how wide a root is cannot change how its arms are paired. In the second of them the chosen pairing joins a 21-pixel arm to a 9-pixel one, so each root halves in width at the pixel where it touches the other, while the pairing 0.12 behind keeps each root at one width and asks it to kink by 20 degrees instead. Which is the better reading is a fair question; the figure is where to ask it.

Where the reading is nearly a coin toss

For the nodes whose gap is narrow, a second figure draws the runner-up: the other pairing, or, where a lateral was left over, the other root it could have been attributed to.

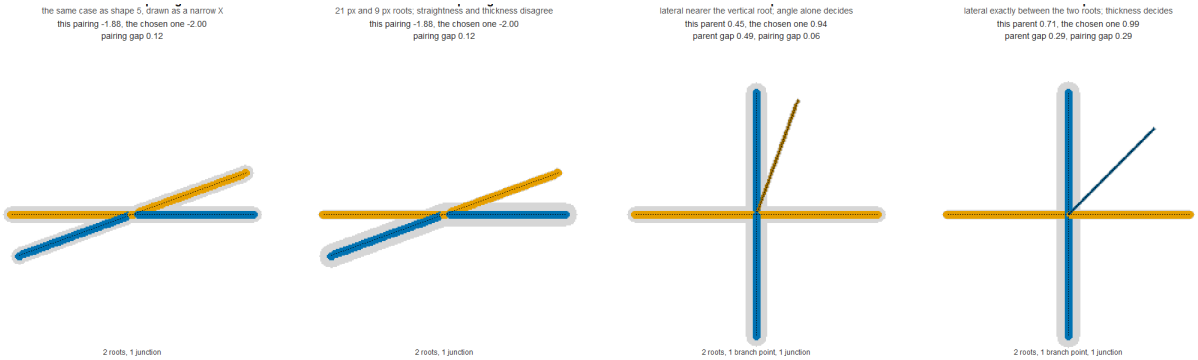


Figure 24: The runner-up reading of four of those nodes.

A lateral left over at a node needs a parent, and the angle does not always give one: in the 45-degree case it ties at 0.71 against both crossing roots. Breaking that tie takes an assumption about where laterals come from — off the thicker axis, or off the thinner higher-order root — and that is a property of the species, not of the image. The package leans towards the thicker arm, through the `dmag` term in `assign_root_order()`.

Two limits worth knowing

Shallow crossings fail in the skeleton, not in the rule. Two one-pixel centerlines crossing at angle t share pixels for roughly $1/\tan t$ of their length either side of the crossing: 1.4 pixels at 20 degrees, 1.9 at 15, 2.8 at 10. Below about 20 degrees that shared run is long enough that the crossing is no longer a single point in the skeleton, and the tracer returns two three-arm junctions a few pixels apart joined by a short segment. Such a node never reaches `resolve_crossings()` at all, and no threshold recovers it. The scoring margin narrows with the angle too — 0.12 at 20 degrees, 0.07 at 15, 0.03 at 10 — but the tracing gives way first.

Preferring the most pairs cuts both ways. It is what reads three roots crossing at one pixel correctly. It can also pair two arms that are not one root: two laterals leaving a node on roughly opposite sides, alike in width, score as well as a thin root crossing through, and the rule takes the pairing because it covers more arms. Nothing in a single node separates those two cases.

Very wide nodes are still left alone. Above eight arms the number of ways to pair them grows faster than the information in them does, and such a node is a tangle rather than a crossing; `verbose = TRUE` reports the nodes where no pair of arms was straight enough to join.

```
data(flatbed_scan_example)
```

```r
seg_fb <- load_flexible_image(flatbed_scan_example, output_format = "spatrast",
                              scale = "binary", select_layer = 2)

# The top edge of the scan carries an artifact rather than root. Anything
# non-root that touches the mask can join otherwise separate roots into one
# component and move the base each path is measured from, so it goes before
# anything is traced.
e <- terra::ext(seg_fb)
seg_fb <- terra::crop(seg_fb, terra::ext(e[1], e[2], e[3],
                                         e[4] - 0.05 * (e[4] - e[3])))

fb_clean <- clean_image(seg_fb,
                        fill_holes_upto   = 50,
                        drop_objects_upto = 50,
                        protect_border    = FALSE)

fb_skl <- skeletonize_image(fb_clean, verbose = FALSE)

# The order map is built on the unpruned skeleton. Pruning removes short
# terminal branches, and which end of the order scale that affects depends on
# the scheme: under order_from_base, used below, terminal segments carry the
# highest numbers, so pruning first flattens the top of the classification.
# `prune_skeleton()` is still the right call before counting tips, and
# `prune_terminal_segments()` prunes inside the graph if you want both.
fb_skl_pruned <- prune_skeleton(fb_skl, mask = fb_clean, min_length = 50,
                                iter = 1)

# The segment graph over a whole tray is the slowest step in this article.
order_res <- branch_order_map(
  skel  = fb_skl,
  mask  = fb_clean,          # the mask is what supplies diameters
  order = "order_from_base",
  unit  = "cm",
  dpi   = 1200
)
#> Cropped to 1900x1964 (offset 0,36)
#> Distance transform...
#> Tracing + contracting junctions...
#>   111 segment(s)
#>   spliced pass-through nodes: 111 -> 111 segment(s)
#>   resolved crossings: 111 -> 111 segment(s)
#> Tip order...
#> Done in 0.36s. 111 segments, max order 8, total length 7857.9 px

# One window for every figure in this part, so the panels can be compared.
fb_frac <- 0.25
fb_ctr  <- c(0.7, 0.625)

# Raw whole, raw magnified, cleaned magnified: the progression that shows what
# cleaning changed rather than asking you to hold two overviews in mind.
zoom_plot(seg_fb,   main = "Segmented, as it comes", frac = fb_frac,
          center = fb_ctr)
```

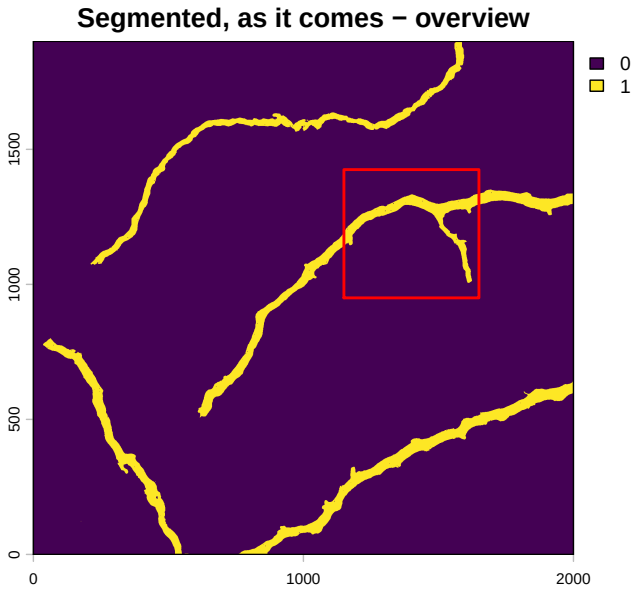


Figure 25: The raw segmentation magnified, specks included.

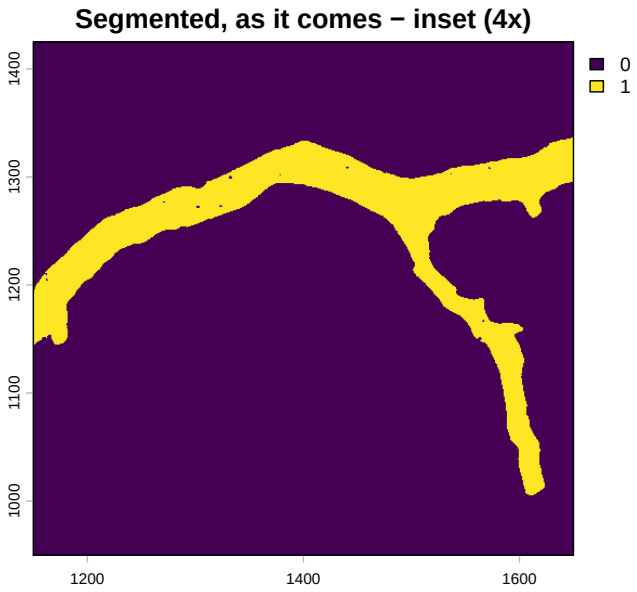


Figure 26: The cleaned scan, with the magnified window outlined.

```
zoom_plot(fb_clean, main = "Cleaned", frac = fb_frac, center = fb_ctr,
          overview = FALSE)
```

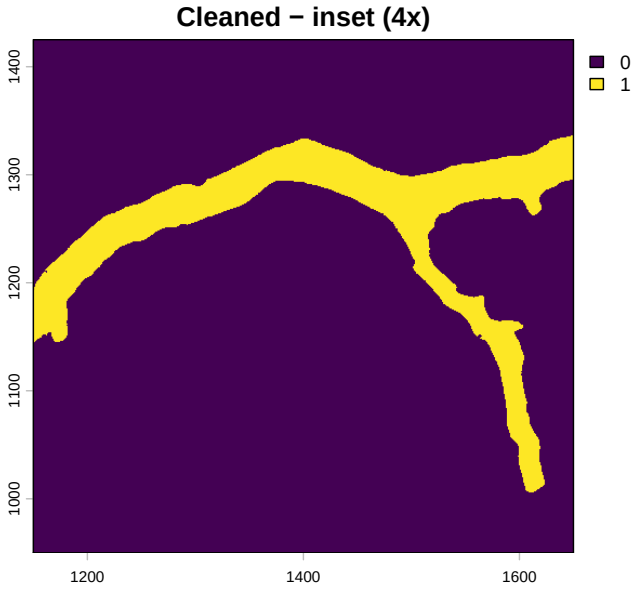


Figure 27: The cleaned scan at native resolution.

```
zoom_plot(fb_skl_pruned, main = "Skeleton (spurs pruned)", frac = fb_frac,
          center = fb_ctr)
```

A one-pixel class map plotted whole is unreadable: the lines are a pixel wide and a continuous palette on a white page gives them almost no contrast. `render_order_overlay()` is the function for looking at it – it paints the per-segment value back onto the **mask**, so each root is drawn at its own width, and picks a labeled swatch per class when the values are a handful of small integers.

```
# render_order_overlay() writes the figure with grDevices::png(), which can
# emit a palette image; terra then reads it as a single band of palette
# indices rather than three color bands. png::readPNG() expands the palette,
# so build the raster from that array and zoom_plot() it like any other image.
as_rgb_rast <- function(file) {
  a <- png::readPNG(file)                          # values in 0-1
  if (length(dim(a)) == 2L) a <- array(a, c(dim(a), 3L))
  r <- terra::rast(a[, , 1:3, drop = FALSE] * 255)
  terra::RGB(r) <- 1:3
  r
}

ord_png <- tempfile(fileext = ".png")
render_order_overlay(attr(order_res$edges, "segments"),
                     order_res$edges$order_from_base,
                     attr(order_res$edges, "dims"), ord_png,
                     mask = fb_clean, value_label = "order from base")
```

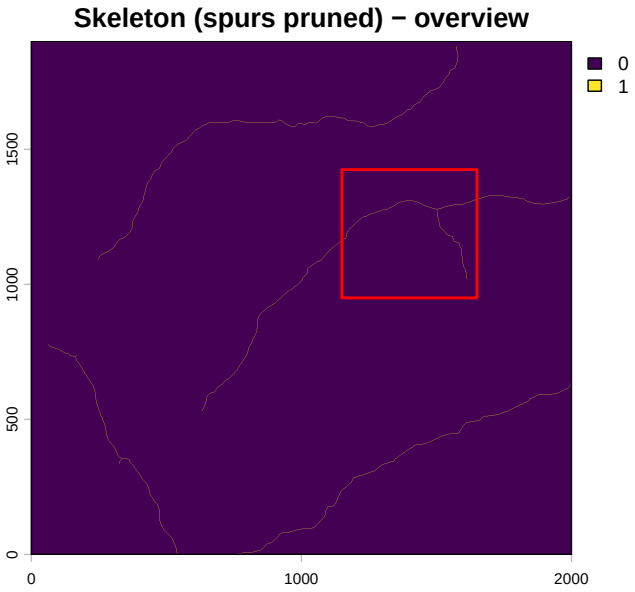


Figure 28: The pruned skeleton of the same scan.

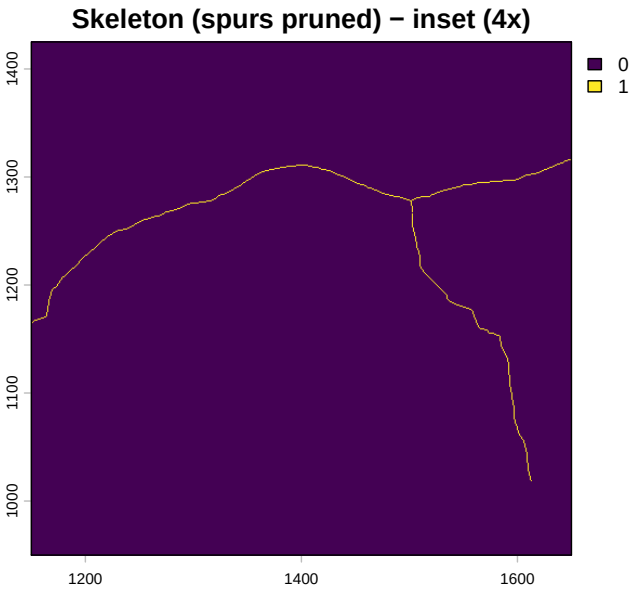


Figure 29: The skeleton at native resolution.

```
# overview = FALSE: the whole tray at page width shows nothing, so only the
# magnified window is drawn.
zoom_plot(as_rgb_rast(ord_png), frac = fb_frac, center = fb_ctr,
          maxcell = Inf, overview = FALSE, main = "Order from base")
```

`plot_order_window()` is the one figure that is not a `zoom_plot()`, because it draws two things that are not in any single raster: the skeleton and, over it, the segments the graph actually traced.

To check the tracing itself rather than the orders, `plot_order_window()` renders a chosen rectangle at native resolution and magnified, with no downsampling at all. It draws the skeleton in gray and the traced, ordered segments in color on top, so **bare gray is skeleton the graph missed** – which is the check to run when a number looks wrong.

```
win_png <- tempfile(fileext = ".png")
plot_order_window(order_res$edges, fb_skl,
                  r_range = c(600, 1000), c_range = c(900, 1300),
                  scale = 2, file = win_png,
                  order_col = "order_from_base")

if (requireNamespace("png", quietly = TRUE)) {
  op <- graphics::par(mar = c(0, 0, 0, 0))
  plot(grDevices::as.raster(png::readPNG(win_png)))
  graphics::par(op)
}
```

All four schemes are computed at once and kept as columns of `order_res$edges`; `order =` only picks which one colors the map and groups the summary. They are not variants of one idea — they count from opposite ends and two of them describe a whole root rather than a segment, so a number means nothing until the scheme is named.

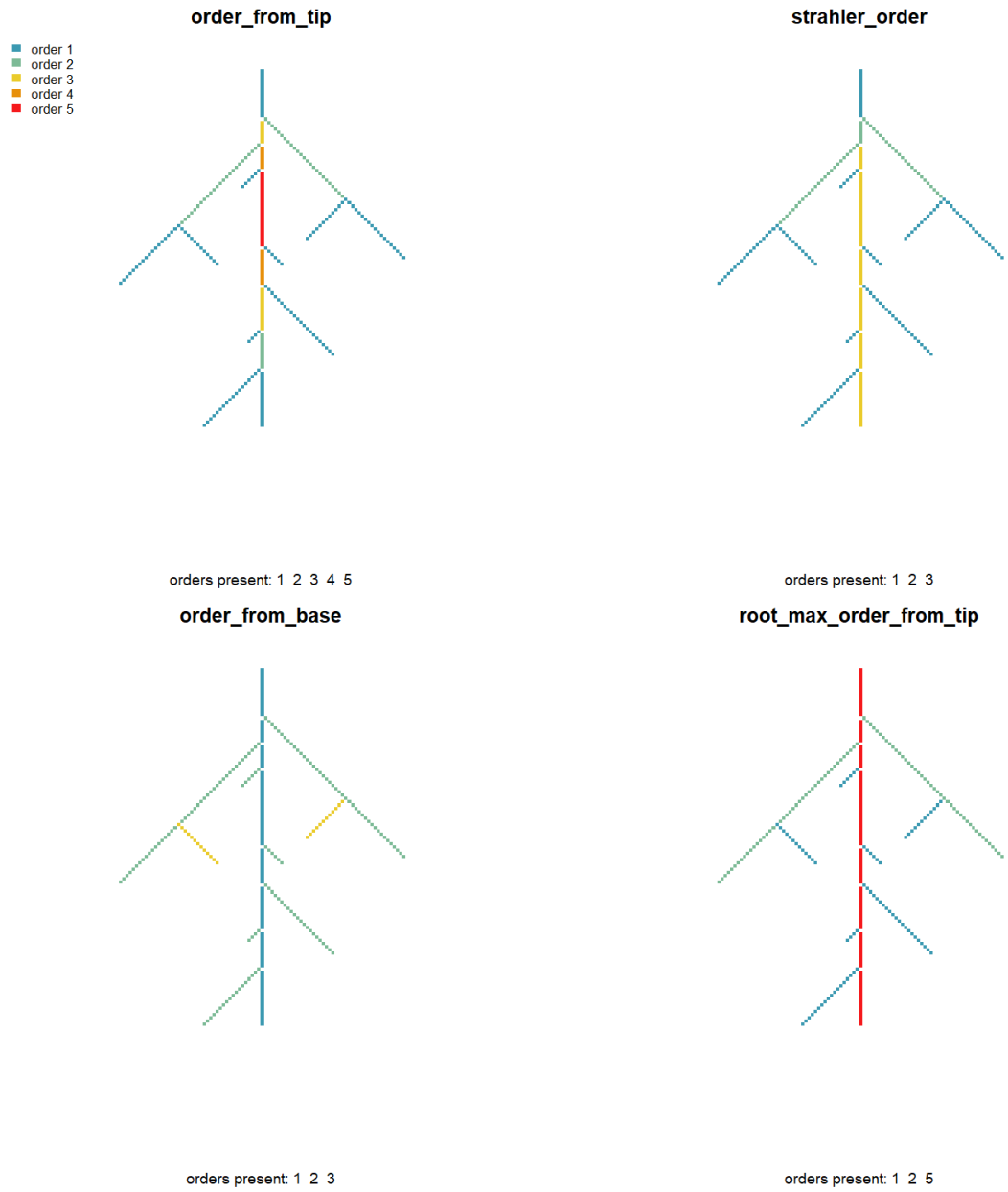
Scheme	Counts from	Unit	1 is
`order_from_tip`	the tips, inward	segment	a terminal segment
`strahler_order`	the tips, inward	segment	a terminal segment
`order_from_base`	the base, outward	continuous root	the main axis
`root_max_order_from_tip`	the tips, inward	continuous root	a root with no branches

`order_from_tip` peels: every terminal segment is 1, and a segment is one more than the highest of its children. `strahler_order` differs in one rule — it increments only where two children of *equal* order meet, otherwise it takes the larger — so it grows with how balanced the branching is rather than with how deep it goes, and stays lower than `order_from_tip` on a lopsided system.

The two per-root schemes first group segments into continuous roots, following the straightest, thickest path through each junction. `order_from_base` counts generations outward, so the main axis of each component is 1, its branches 2, theirs 3. `root_max_order_from_tip` stamps every segment of a root with the largest `order_from_tip` found along it, which labels a root by how much branching it carries rather than by where it sits.

Which to use follows from the question. Comparing against published root-order data usually means `order_from_base`, because that is the convention most root ecology reports. Asking how deep the branching hierarchy goes means `order_from_tip`. Asking how *balanced* it is means `strahler_order`. The one thing not to do is compare numbers across schemes.

A real scan is too dense to show that, so the four schemes are drawn on one synthetic fan instead: a trunk with laterals leaving at six heights, two of them carrying their own pair of children, and four stubs. Nine branch points, eleven tips, three levels deep.



The same stub carries a different number in every panel, which is the point: the schemes are not refinements of one another.

```
order_res$summary
#>   order n_segments n_tips n_branch_points total_length cumulative_length cumulative_fraction mean_se
#> 1     1         26      6              22   5.16391097          5.163911           0.3247345
#> 2     2         38     25              13   3.30586934          8.469780           0.5326253
#> 3     3         27     15              14   4.75573400         13.225514           0.8316914
#> 4     4         14     12               2   0.71618856         13.941703           0.8767292
#> 5     5          4      2               2   1.91089531         15.852598           0.9968966
#> 6     6          2      2               0   0.04935045         15.901949           1.0000000
#>   branching_frequency mean_diameter median_diameter
#> 1            4.260337    0.06893584      0.06845654
#> 2            3.932400    0.04545475      0.04095795
```

```
#> 3            2.943815    0.06355430      0.04190786
#> 4            2.792561    0.04367593      0.03722479
#> 5            1.046630    0.04563837      0.03825043
#> 6            0.000000    0.03223201      0.03160465

# The thickest order class against everything else, picked by diameter rather
# than by order number, so the same call means the same thing across a dataset.
order_metrics(order_res, groups = "thickest")
#>      group    orders n_segments n_tips n_branch_points total_length length_fraction mean_segment_len
#> 1 thickest         1         26      6              22     5.163911       0.3247345           0.1986
#> 2     rest 2,3,4,5,6         85     56              31    10.738038       0.6752655           0.1263
#>   branching_frequency mean_diameter median_diameter
#> 1            4.260337    0.06893584      0.06845654
#> 2            2.886933    0.05332406      0.03993719
```

Tips and branch points come from this table too: `n_tips` and `n_branch_points` per segment, summed per order class in `$summary`.

```
diam_fb   <- root_diameter(fb_clean, skeleton_img = fb_skl_pruned,
                           unit = "cm", dpi = 1200, select_layer = NULL)
diam_vals <- diam_fb$diameters      # the per-skeleton-pixel values

cat(sprintf("diameter mean %.4f cm | sd %.4f | max %.4f cm\n",
            mean(diam_vals), stats::sd(diam_vals), max(diam_vals)))
#> diameter mean 0.0618 cm | sd 0.0164 | max 0.1108 cm

ggplot(data.frame(diameter = diam_vals), aes(x = diameter)) +
  geom_histogram(bins = 24, fill = "steelblue4", color = "white") +
  theme_minimal() +
  labs(x = "Diameter (cm)", y = "Count",
       title = "Root diameter distribution")
```

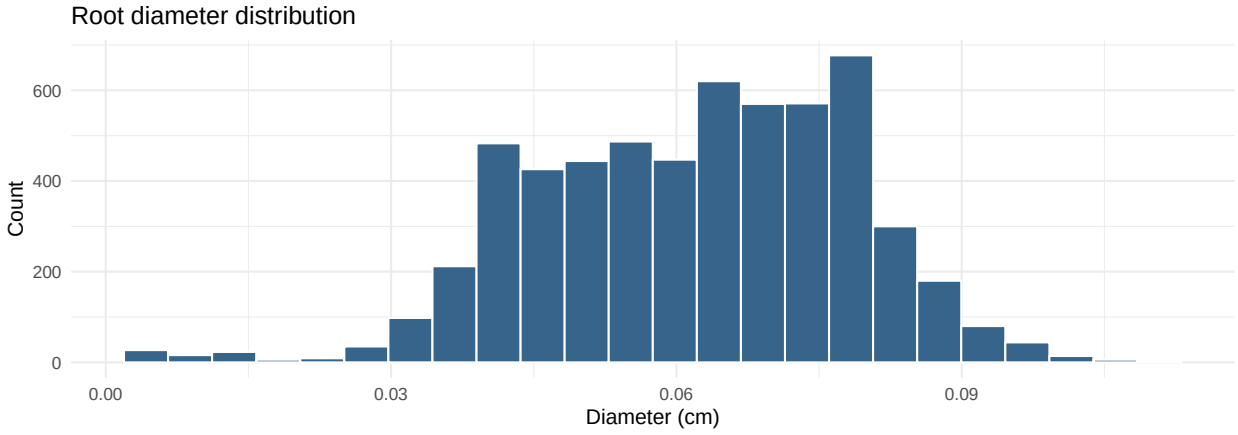


Figure 30: Root diameters across the flatbed window.

A diameter distribution is where fine and coarse roots separate, and `modal_peaks()` finds the modes in it, optionally refining them with a Gaussian mixture via `mclust`. It is not run here because this one distribution is unimodal: a single peak with a long upper tail, which is what a scan of mostly fine roots looks like. Run it where the histogram shows two humps and you want the cut between them placed by the data rather than by eye.

Network shape and transport

Two questions the segment table answers once it exists, both working on `order_res$edges` rather than on the image.

```
net <- network_metrics(order_res$edges, topology = "projected_tree")
net$traits
#> $n_nodes
#> [1] 115
#>
#> $n_edges
#> [1] 111
#>
#> $n_components
#> [1] 4
#>
#> $n_tips
#> [1] 62
#>
#> $n_branch_points
#> [1] 53
#>
#> $tip_ratio
#> [1] 0.5391304
#>
#> $beta
#> [1] 0.9652174
#>
#> $total_length
#> [1] 15.90195
#>
#> $mean_edge_length
#> [1] 0.1432608
net$qc
#> $cyclomatic
#> [1] 0
#>
#> $alpha_meshedness
#> [1] 0
#>
#> $gamma
#> [1] 0.3274336
#>
#> $n_self_loops
#> [1] 0
#>
#> $n_duplicate_edges
#> [1] 0
#>
#> $is_forest
#> [1] TRUE
#>
#> $artefact_cycles
#> [1] 0
```

```
#>
#> $artefact_rate
#> [1] 0
net$note
#> [1] "loops in a projected tree are crossings the segmentation did not separate; artefact_rate divide
```

`$traits` is the shape of the network: node and edge counts, how many separate pieces, how many tips, the tip ratio, and `beta` – edges per node, which is the plainest measure of how branchy a network is. `$qc` holds the loop arithmetic. The same numbers land in one place or the other depending on the `topology` you declare, and nothing else changes:

`topology`	What a loop means	Where loop counts go
`"tree"`	impossible	`$qc`: any loop means the input is not a tree
`"projected_tree"`	crossings in projection	`$qc`: unresolved overlaps
`"reticulate"`	real, e.g. mycelium	`$traits`: a measurement

On a root scan a closed circuit is almost never one the plant grew; it is two roots crossing in front of one another, touching in the image, joined by the skeleton. So `artefact_cycles` counts them and `artefact_rate` divides by total network length, which makes the count comparable between a large root system and a small one. Reporting that as biology is the mistake the argument exists to prevent – which is also why `"reticulate"` is there: for mycelium with anastomoses, leaf venation or a stream network, the loops *are* the structure, and `cyclomatic` and `alpha_meshedness` move into `$traits`.

```
et <- root_transport(order_res$edges)
head(et[, c("length", "path_length", "path_resistance",
            "downstream_tips", "downstream_length")], 4)
#>       length path_length path_resistance downstream_tips downstream_length
#> 1 0.04264472    2.221137        6827712               1        0.04264472
#> 2 0.04009685    2.218590        6929970               1        0.04009685
#> 3 0.18486079    2.178493        6337904               2        0.26760236
#> 4 0.04135668    2.034989        6777683               1        0.04135668
```

Everything here answers one question: which parts of this root system are *remote*, and remote in what sense. Geometric distance from the base is one answer and hydraulic cost is another, and they disagree, because a root's resistance scales with the fourth power of its radius. A tip at the end of a long thick root can be hydraulically closer to the base than one at the end of a short thin root. That is the comparison the figure below draws, and it is why both columns are reported.

Two families of column, and the direction of each is fixed by the base rather than by any order scheme. `root_transport()` picks the thickest loose end of each separate piece as the base — on a scan that is the proximal end — and everything is measured relative to that. `path_length` and `path_resistance` accumulate **inward to outward**: from the base to the far end of this segment, so they grow toward the tips and answer "how far is this, from the plant's point of view". `downstream_tips`, `downstream_length` and `downstream_surface` accumulate **outward to inward**: what lies beyond this segment, so they peak at the base and answer "how much does this segment serve". Neither changes if you switch `order =` in `branch_order_map()`; both change if the base is picked wrongly, which is what a stray non-root object touching the mask does. Resistance is the Hagen-Poiseuille term for an open pipe, so it scales with the fourth power of radius: halving a root's diameter makes the same length roughly sixteen times harder to push water through. Real resistance is set by xylem anatomy, not by the outer diameter, so treat these as a relative ranking within one scan rather than a physical measurement.

`render_order_overlay()` paints any per-segment column back onto the mask, which is how a transport quantity becomes legible. It picks its own legend: a handful of small integers gets one labeled swatch per class, anything continuous gets a color ramp with a scale bar.

```
png_file <- tempfile(fileext = ".png")

# max_side caps the longest side of the written PNG, and anything below the
# scan's own width throws pixels away: thin roots drop out rather than getting
# thicker. Keep it at the full width and zoom afterwards instead.
# A handful of thin distal segments carry resistance far above the rest, and a
# ramp stretched to that maximum leaves every other segment in nearly the same
# color. Capping the value at the 97th percentile spends the ramp on the range
# the roots actually occupy; the capped segments sit at the top color. The
# alternative is to paint log10(path_resistance) and label the bar as a log
# scale, which keeps the extremes distinguishable but makes the bar harder to
# read as a physical quantity.
cap <- stats::quantile(et$path_resistance, 0.97, na.rm = TRUE)

render_order_overlay(attr(et, "segments"),
                     pmin(et$path_resistance, cap),
                     attr(et, "dims"), png_file,
                     mask = fb_clean, max_side = 2000,
                     value_label = "path resistance (capped at p97)")

zoom_plot(as_rgb_rast(png_file), frac = fb_frac, center = fb_ctr,
          maxcell = Inf, overview = FALSE, main = "Path resistance")
```

Segments with no value are drawn in red, which makes failures visible rather than silently absent – a loop in the graph has no single "beyond", so its transport columns are `NA` and it shows up in that color. A lot of red is the same signal `network_metrics()` reports as `artefact_cycles`.

Path length and path resistance rank the segments differently, and the reason is the radius term.

```
op     <- graphics::par(mfrow = c(1, 2), mar = c(4, 4, 3, 1))
pt_bg  <- grDevices::rgb(0.21, 0.35, 0.47, 0.45)
pt_cex <- 0.4 + 2.4 * et$mean_diameter / max(et$mean_diameter, na.rm = TRUE)

plot(et$path_length, et$path_resistance,
     pch = 21, bg = pt_bg, col = "steelblue4", cex = pt_cex,
     xlab = "path length from the base (cm)",
     ylab = "path resistance (relative)",
     main = "linear resistance axis")

plot(et$path_length, et$path_resistance, log = "y",
     pch = 21, bg = pt_bg, col = "steelblue4", cex = pt_cex,
     xlab = "path length from the base (cm)",
     ylab = "path resistance (relative, log)",
     main = "log resistance axis")
```

```
graphics::par(op)
```

The pair is worth seeing, and which one to use depends on the scan. On the linear axis a few thin segments set the top of the range and everything else is squeezed into a band along the bottom, which says only that

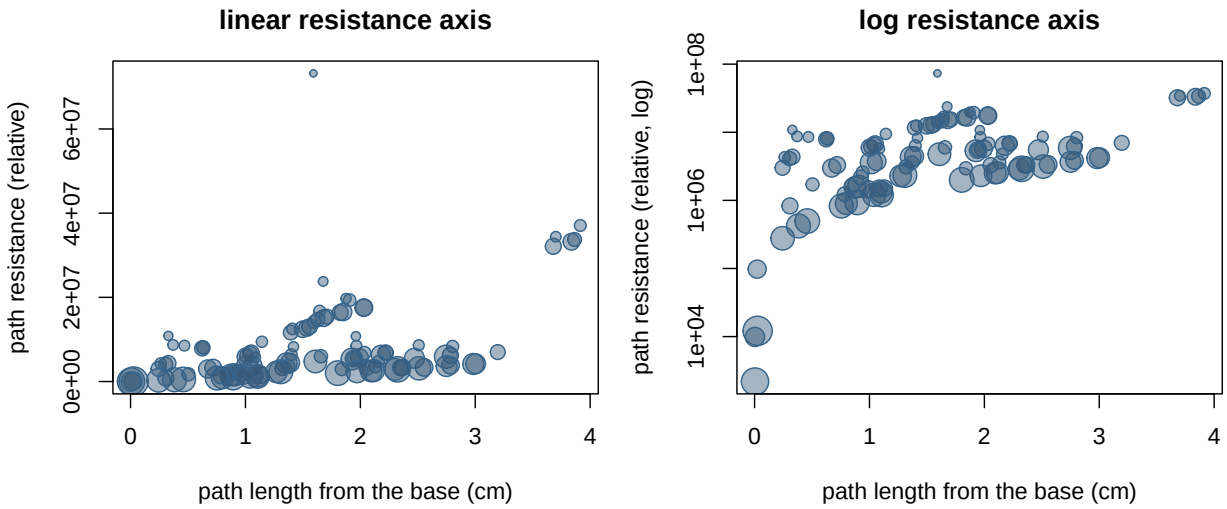


Figure 31: Path resistance against path length.

most segments are cheap and one or two are not. The log axis spreads the same points over the orders of magnitude they actually span, and the structure appears: at any given distance from the base, resistance varies by two or three orders of magnitude. A clean scan with no stray object can be readable on the linear axis; one where a single artifact has created an extremely thin segment never is, because that segment alone sets the scale. Removing the artifact is the better fix, and the log axis is what to reach for when you cannot.

Point size is diameter, and the small points – the thin roots – sit at the top. So a long thick root can be an easier path for water than a short thin one, which is why both columns are reported rather than one standing in for the other.

Reference

What you can measure

One row per quantity, with the unit it comes back in and what it says. Lengths, diameters and depths follow `unit` and `dpi`; the table assumes `"cm"`.

Per image, or per depth interval

Quantity	Unit	Meaning
`rootpx`, `voidpx`	pixels	root and background pixel counts
`rootpx.density`	%	root pixels as a share of the interval
`rootlength`	cm	skeleton length, Kimura estimator
`rootlength.density`	cm cm^-2	length per unit of imaged wall
`avg.diameter`, `max.diameter`	cm	from 2x the distance transform
`var.diameter`	cm^2	spread of diameter within the interval
`root.surface.area`	cm^2	lateral area of the root cylinder
`root.volume`	cm^3	volume of the same cylinder
`n.diameter.peaks`	count	modes in the diameter distribution
`rcc`, `gcc`, `bcc`	fraction	chromatic coordinates, sum to 1
`hue`, `saturation`	0-1	HSV of the mean color
`luminosity`	0-255	Rec. 709 weighted mean

Per tube or scan

Quantity	Unit	Meaning
`mrd`	cm	mean rooting depth, amount-weighted
`total.length.density`	cm cm^-2	summed over the profile
`rootlength.fraction`	fraction	share of total length in an interval
`deep_drive`	0-1	share of root pixels heading deepest
`mean.steepness.angle`	degrees	0 horizontal, 90 vertical
Wasserstein distance	cm	depth the mass must travel between profiles
Jensen-Shannon, KL	nats	bin-by-bin divergence, no depth metric
`amplitude`, `offset`	trait units	circumferential sine fit
`peakTime`	slice index	where that sine peaks around the tube

Per segment, from the branching graph

Quantity	Unit	Meaning
`length`	cm	along the pixel chain, junction stub included

Quantity	Unit	Meaning
`mean_diameter`	cm	length-weighted within the segment
`order_from_tip`, `strahler_order`	count	inward from the tips, per segment
`order_from_base`	count	generations outward, per root
`root_max_order_from_tip`	count	the root's largest tip order
`n_tips`, `n_branch_points`	count	degree-1 and degree-3+ nodes
`branching_frequency`	cmˆ-1	branch points per unit length
`path_length`	cm	base of the root system to this segment
`path_resistance`	relative	8 eta L / (pi rˆ4), eta = 1
`downstream_length`	cm	root length beyond this segment
`downstream_tips`	count	tips beyond it
`downstream_surface`	cmˆ2	root surface beyond it

Per network

Quantity	Unit	Meaning
`n_nodes`, `n_edges`	count	graph size
`n_components`	count	separate root systems in the image
`tip_ratio`	fraction	tips over nodes
`beta`	edges/node	how branchy the network is
`cyclomatic`	count	independent closed circuits
`alpha_meshedness`	0-1	loops against the planar maximum
`artefact_cycles`	count	the same, read as unresolved crossings
`artefact_rate`	cmˆ-1	crossings per unit length, comparable

Conventions

Four things are read the same way everywhere, and they are the ones to check before trusting a number.

- **Pixels to physical:** `cm = px * 2.54 / dpi`. Every function that reports a length needs `dpi`, and `root_diameter()` defaults to 300 rather than erroring.
- **Angles:** `angle_from_vertical` and `insertion_angles` are degrees from vertical, `[0, 90]`, 0 upright. `mean.steepness.angle` is degrees from horizontal. They are not the same convention.
- **Axes:** depth runs along the image width, the tube circumference down the height. A scan with the soil surface at the top has to be turned a quarter turn first.
- **Depth zero:** `start_soil` and `soil_starts` are centimeters, subtracted from every depth. `estimate_soil_surface()` reports pixels, so convert.

Two more are easy to misread. Diameter in the order summaries is length-weighted, so `mean_diameter` describes the diameter of the root length rather than the average of segment values. And the Kimura length estimators are not additive, so depth intervals do not sum to the whole-image value.

Packages this builds on

Rootopia is mostly glue and geometry. The numerical work is done by other packages, and knowing which one is behind a function tells you where to look when a result surprises you.

Package	What Rootopia uses it for	Required?
terra	every raster: reading, extents, `zonal()`, plotting	yes
imager	connected components, morphology, the distance transform behind `root_diameter()`	yes
transport	`wasserstein1d()`, the earth-mover distance in `compare_depth_distribution()`	yes
minpack.lm	Levenberg-Marquardt fit in `fit_sine_curve()` / `rhythmicity()`	yes
mclust	Gaussian mixture clustering in `modal_peaks()`	yes
glcm	the texture matrices in `analyze_soil_texture()`	yes
imagefx	phase and cross correlation in `estimate_rotation_shift()`	yes
raster	the object type glcm and RStoolbox expect	yes
dplyr, **tidyr**, **stringr**	shaping the per-interval tables the wrappers return	yes
landscapemetrics	every metric in `root_scape_metrics()`	suggested
RStoolbox	unsupervised classification in `estimate_soil_surface()` and `estimate_rotation_center()`	suggested
png, **tiff**	reading and writing those formats in stitching and overlays	suggested
ggplot2	figures in the vignettes only	suggested

A suggested package is not installed with Rootopia. The functions that need one check for it and stop with a message naming it rather than failing further in, so a thin install still runs the depth-profile workflow in Part I.

The method for `rhythmicity()` is not original here either: it reproduces the likelihood-based rhythmicity tests of diffCircadian (Ding *et al.* 2021) for a spatial axis instead of a temporal one.

Availability

Rootopia is an R package, available from `https://github.com/jcunow/Rootopia` and not yet on CRAN. It requires R >= 4.1 for the native pipe and `terra` >= 1.8. `help(package = "Rootopia")` lists every exported function with its arguments and return value, and further worked examples ship with the package.

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Software

Entries as `citation()` prints them, with the package versions used here. Where a package names a paper or a book instead of its manual, that is what is cited.

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